

Topic 5.1
Supervised Learning: Tree-Based Models

Introduction to Tree-Structured Models

By Dr Lee How Chinh

Objectives

- Describe and illustrate tree-structured models

Decision Trees

- Decision trees are statistical models designed for **supervised prediction problems**.
- So we need a set of **input variables (predictors)** to predict the value of one or more **target variables (outcome)** in Decision Tree predictions.
- The data used to estimate a Decision Tree model is a set of **cases (observations, examples)** consisting values of inputs and target.

Decision Trees

- Then the fitted Decision Tree model is usually applied to new cases where the target is unknown

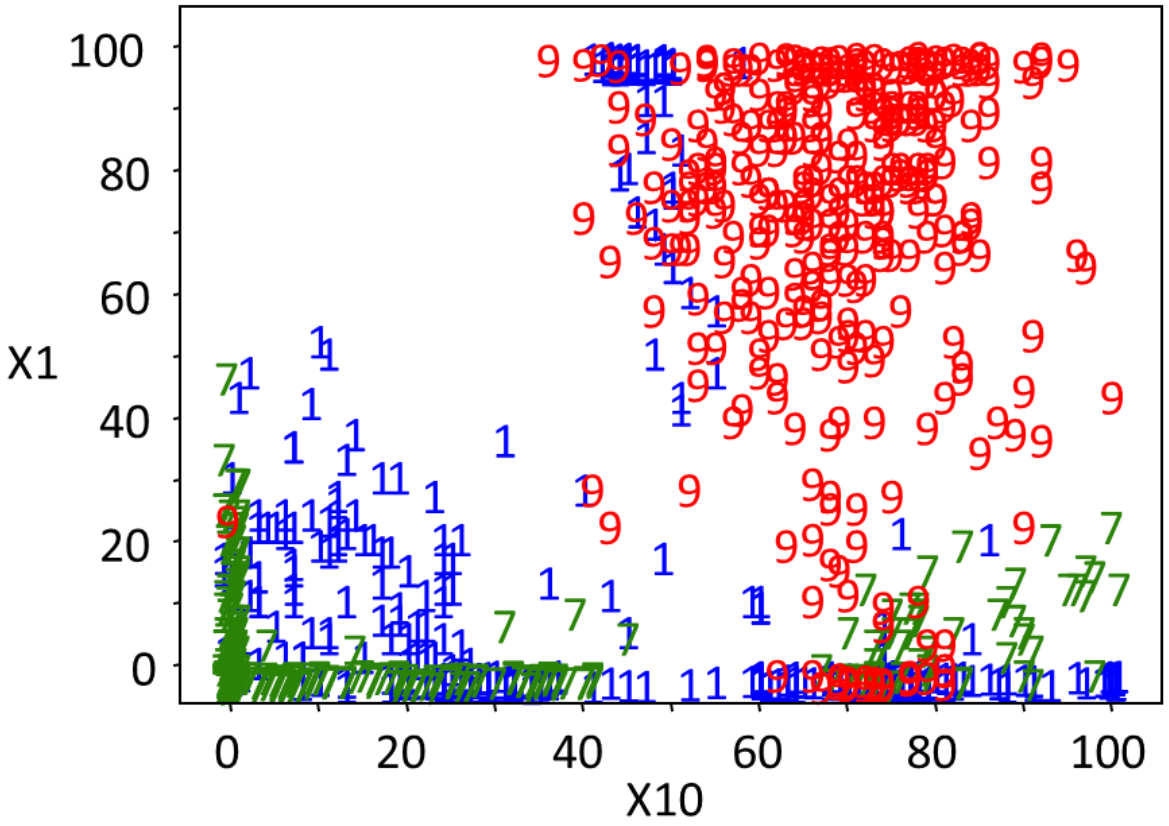
- The target variables of a Decision Tree model can be Categorical or Numeric.

Decision Tree Supervised Prediction: Nominal Target

- Handwriting recognition is a classic example of supervised prediction.
- The example data set is a subset of the pen-based recognition of handwritten digits data. (Data available from the UCI repository, Blake et al. 1998)
- The cases are digits written on a pressure-sensitive tablet.
- The inputs variables measure the position of the pen.
- They are scaled from 0 to 100.
- The target is the true written digit (0-9)

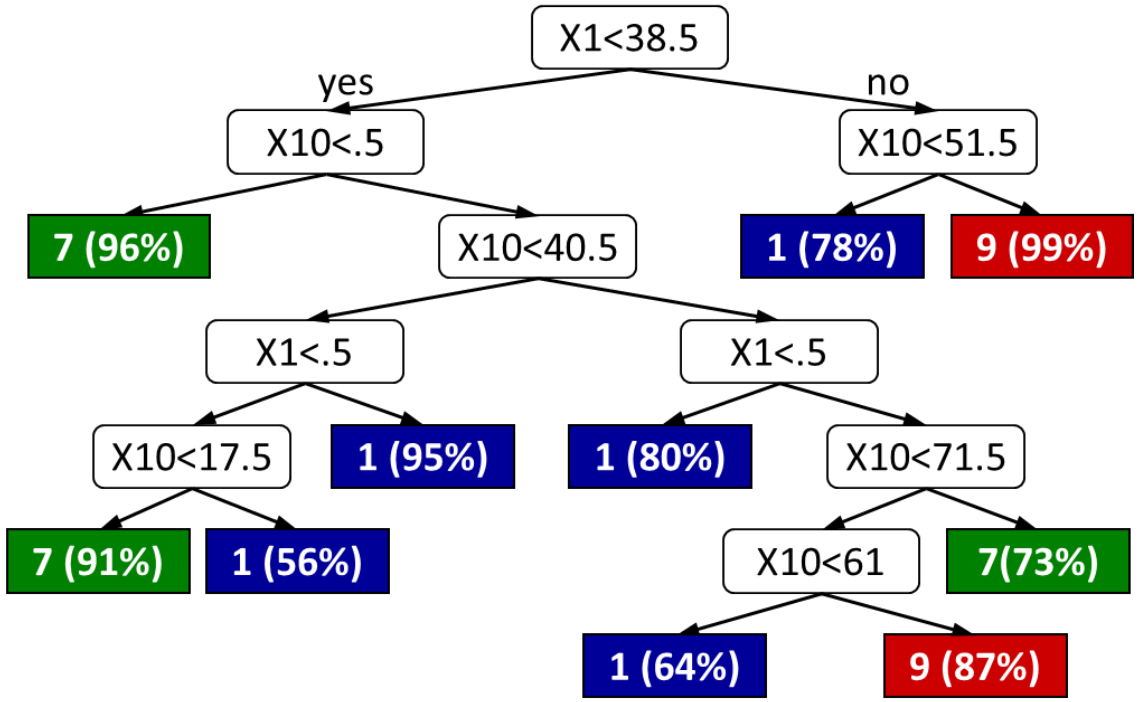
Decision Tree Supervised Prediction: Nominal Target

- Two of the original 16 inputs, **X1** and **X10**, are shown.
- This subset contains the 1064 cases corresponding to the three digits **1, 7** and **9.**
- In this data, the inputs have an ratio measurement scale and the target has an nominal measurement scale.



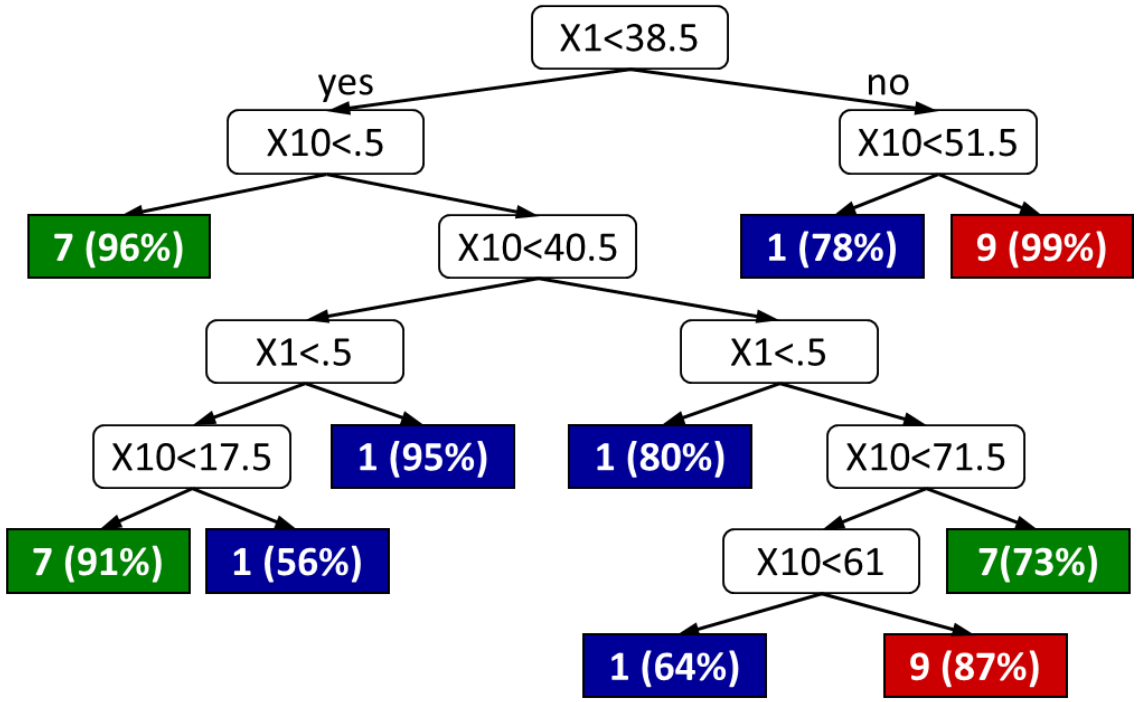
Decision Tree Supervised Prediction: Nominal Target

- The name of decision tree because the predictive model can be represented in a tree-like structure.
- A decision tree is read from the top starting at the **root node**
- Each internal node represents a split based on the values of one of the inputs.
- Cases move down the branch that contains its input value.



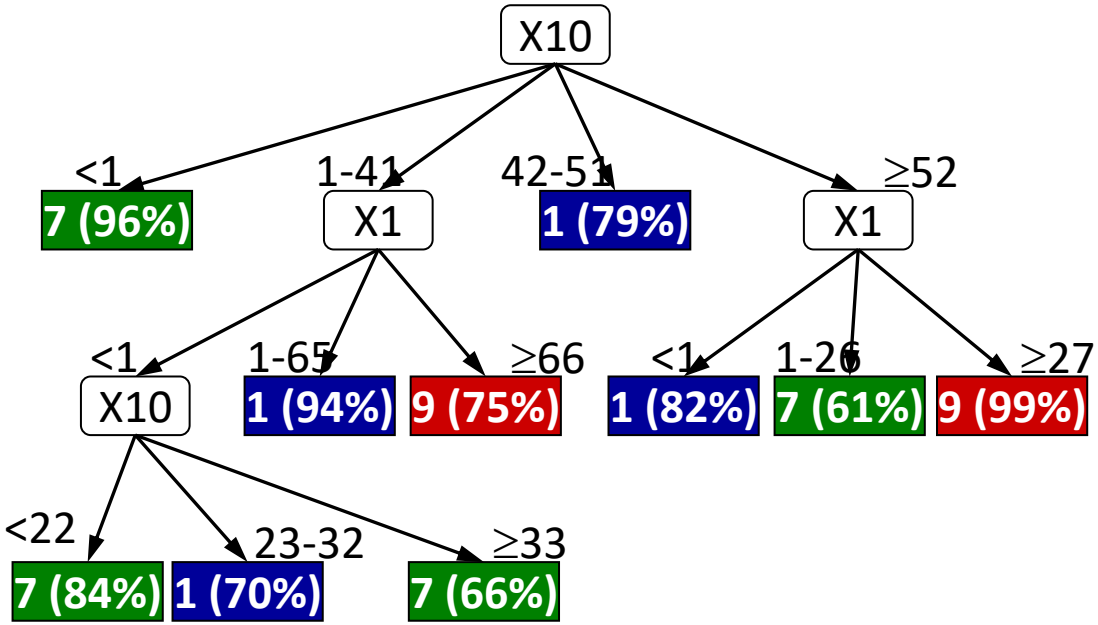
Decision Tree Supervised Prediction: Nominal Target

- In a binary tree with interval inputs, each internal node is a simple inequality.
- A case moves left if the inequality is true and right otherwise.
- The terminal node of the tree are called **leaves**.
- The **leaves** represent the predicted target.

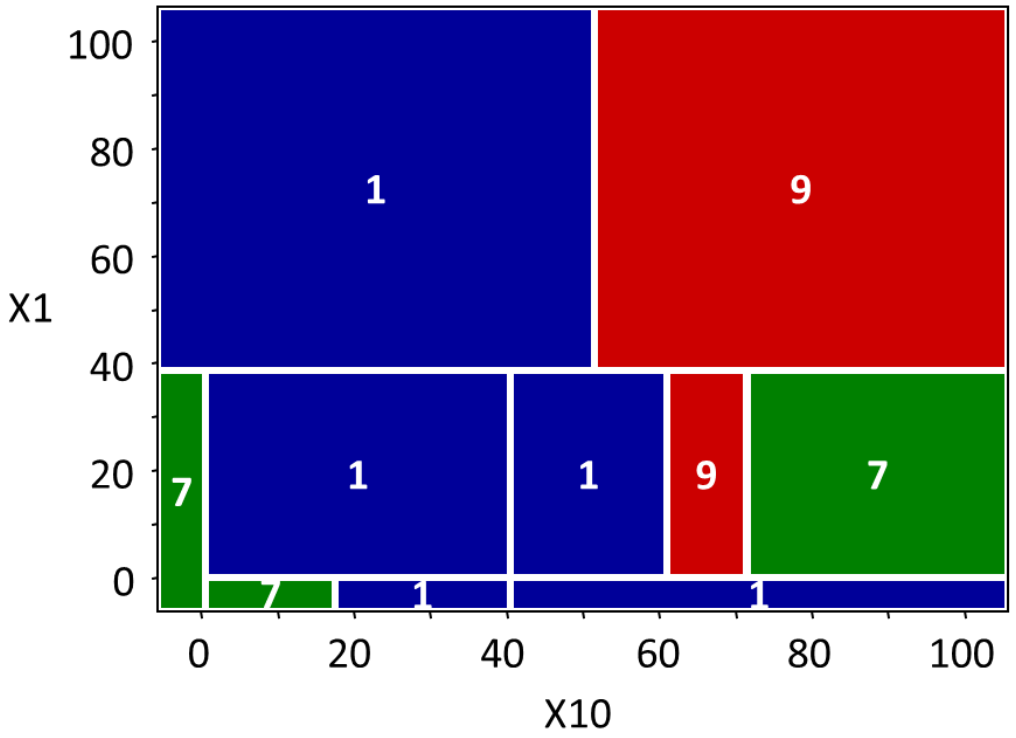


Decision Tree Supervised Prediction: Nominal Target

- When the target is categorical, the model is called a **classification tree.**
- Decision trees can have multi-way splits where the values of the inputs are partitioned into disjoint ranges.
- The leaves give the predicted class and the probability of that class.



Decision Tree Supervised Prediction: Nominal Target



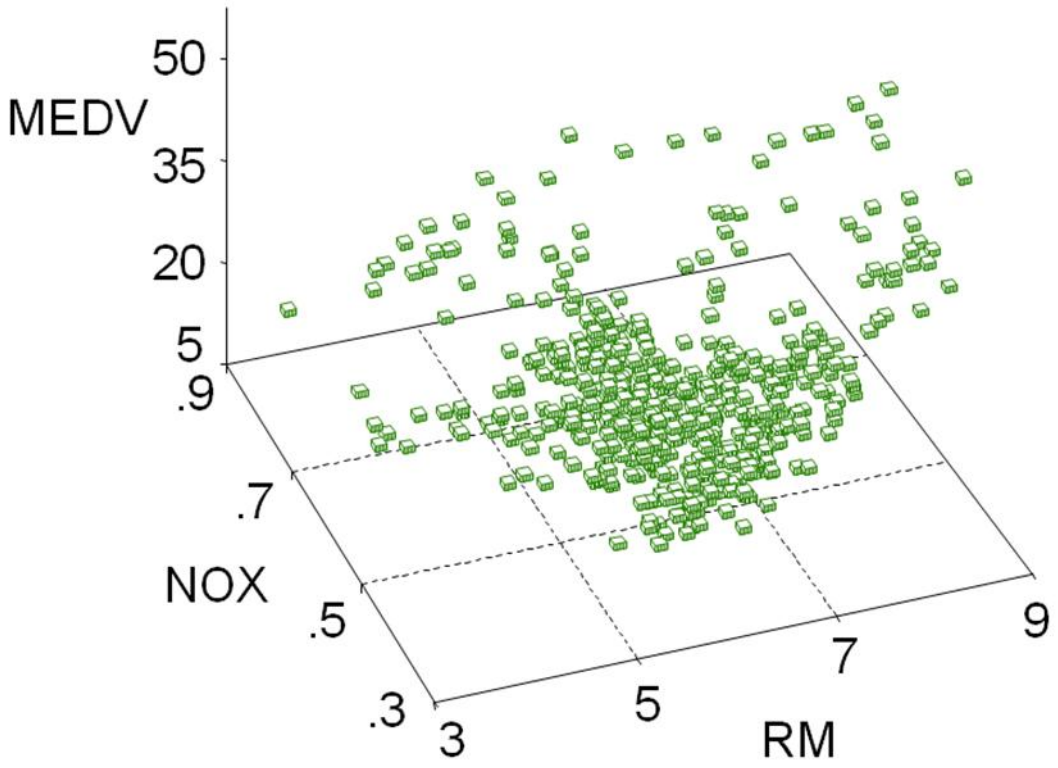
Leaf	Pr(1|x)	Pr(7|x)	Pr(9|x)	Decision
1	.03	.96	.01	7
2	.09	.91	.00	7
3	.56	.44	.00	1
4	.95	.05	.00	1
5	.80	.10	.10	1
6	.64	.09	.27	1
7	.00	.13	.87	9
8	.10	.73	.17	7
9	.78	.01	.21	1
10	.01	.00	.99	9

The leaves of a classification tree split the input space into rectilinear regions.

The decision of each class is determined by the posterior probability of that class.

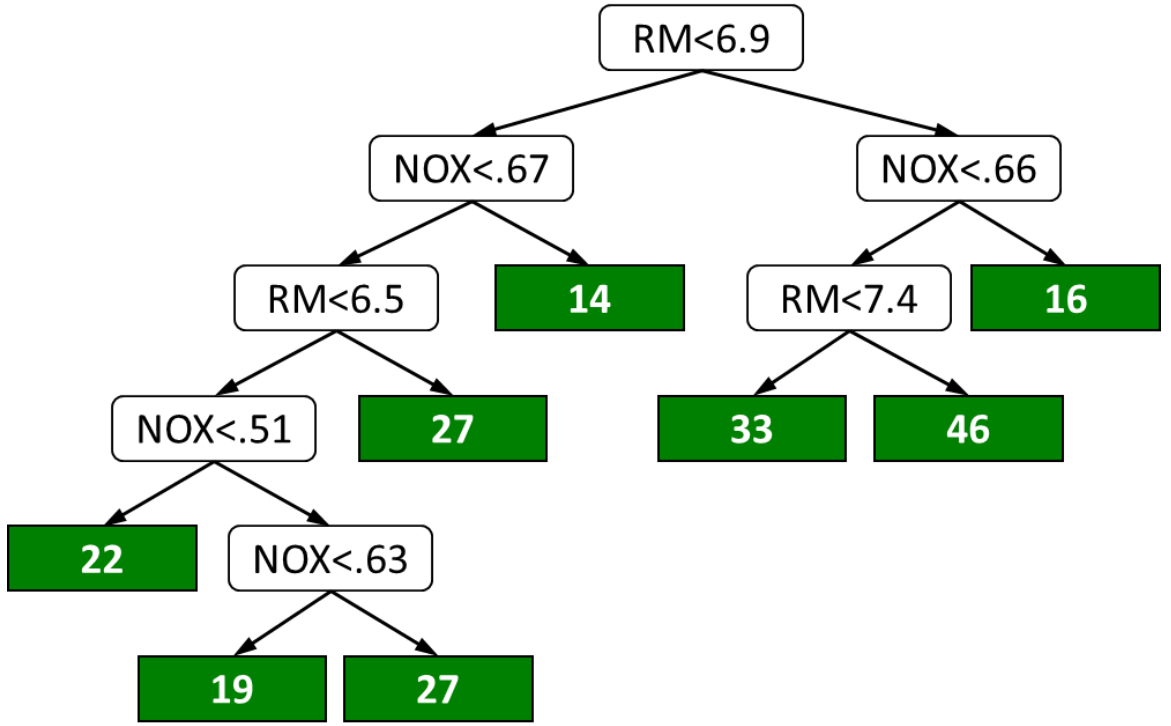
Decision Tree Supervised Prediction: Interval Target

- The Boston housing data with 506 cases that census in Boston. (Data available from the UCI repository, Blake et al. 1998)
- The target is median home value (**MEDV**)
- Two of the 13 inputs are shown:
  - the average number of rooms (**RM**)
  - the nitrogen oxide concentration in the air (**NOX**)



Decision Tree Supervised Prediction: Interval Target

- When the target is numeric continuous, the model is called a **regression tree**.
- The leaves provide the predicted value of the target.
- All cases that reach a particular leaf are assigned the same predicted value.



Decision Tree Supervised Prediction: Interval Target

- The path to each leaf can be expressed as a Boolean rule.

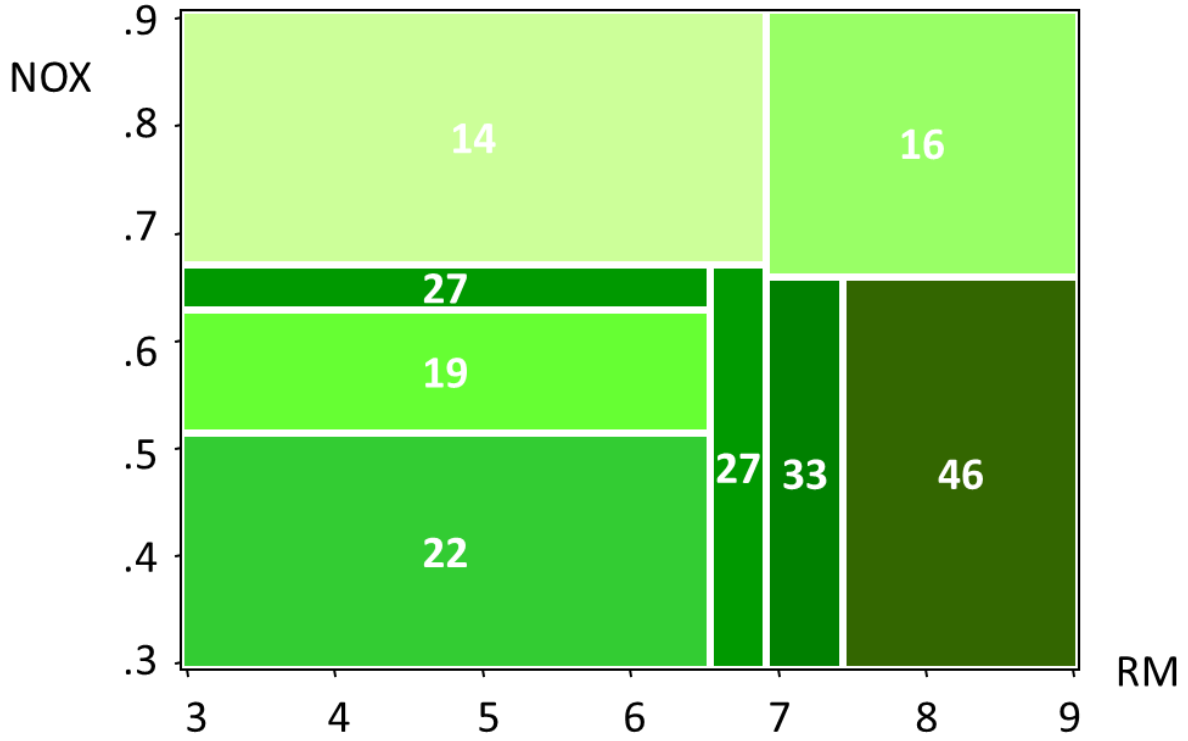
If the inputs $\in$ {region of the input space}, then the predicted value $=$ *value*.

If RM $\in$ {*values*} and NOX $\in$ {*values*}, then MEDV=*value*.

Leaf	RM	NOX	Predicted MEDV
1	<6.5	<.51	22
2	<6.5	[.51, .63)	19
3	<6.5	[.63, .67)	27
4	[6.5, 6.9)	<.67	27
5	<6.9	≥.67	14
6	[6.9, 7.4)	<.66	33
7	≥7.4	<.66	46
8	≥6.9	≥.66	16

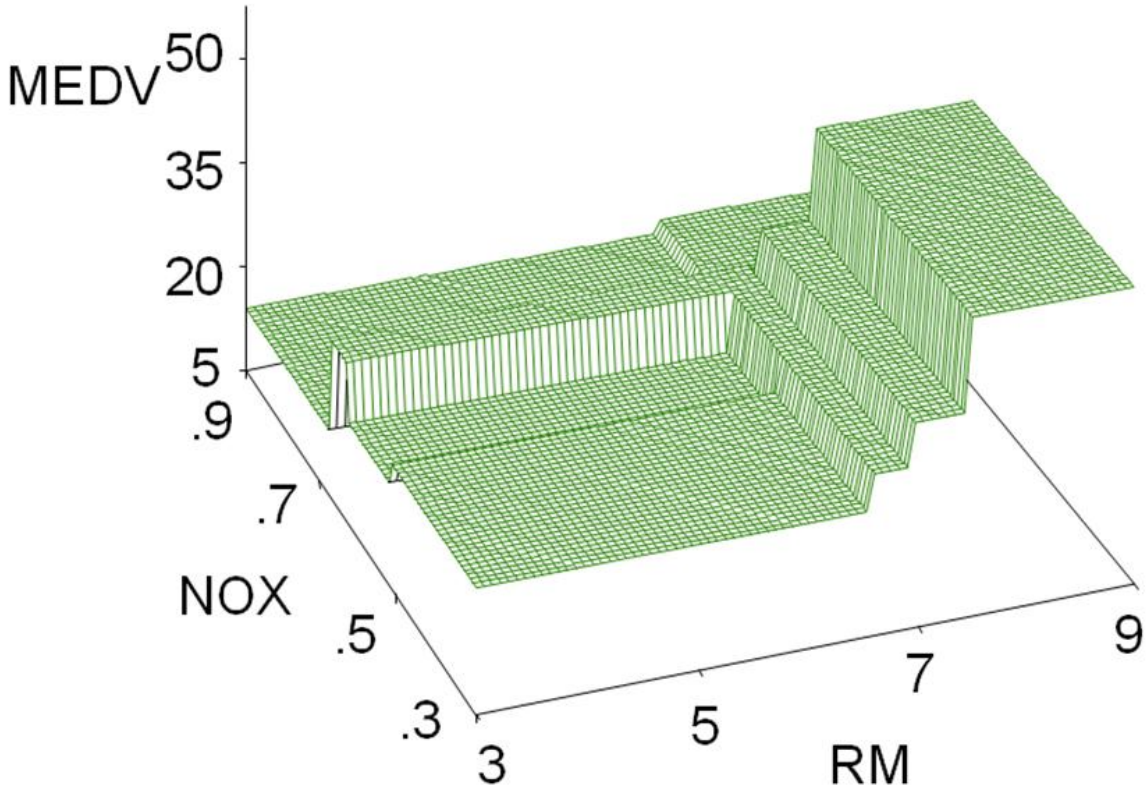
Decision Tree Supervised Prediction: Interval Target

- The leaves of the decision tree partition the input space into rectilinear regions.
- Each partition contains a predicted target value.



Decision Tree Supervised Prediction: Interval Target

- The fitted regression model is a multivariate step function.
- The surface is a piecewise constant and not joined continuously at the boundaries.
- A **step function** is highly flexible and capable of modelling nonlinear patterns.
- Therefore, decision trees are known as **universal estimators.**



Lesson Summary

- A decision tree starts from the root node, and each internal node splits based on the values of one input, then the tree ends with the terminal node called leaves. Each branch contains the cases that agree with the value of the inputs.
- A decision tree with categorical targets is called a classification tree, and with continuous targets is called a regression tree
- The path of each leaf can be expressed as a Boolean rule.
- A regression tree is a universal estimator because it is a highly flexible multivariate step function.

How does a decision tree select useful inputs?



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Topic 5.2
Supervised Learning: Tree-Based Models

Selecting useful inputs: recursive partitioning

By Dr Lee How Chinh



Objectives

- Describe and illustrate recursive partitioning
- Describe the combinatorics involved in split searches
- Describe the splitting criteria
- Explain how missing values are treated in decision trees

Contents

- Recursive Partitioning
- Split Search
- Splitting Criteria
- Missing Values
- Variable Importance
- Interval Targets

Recursive Partitioning

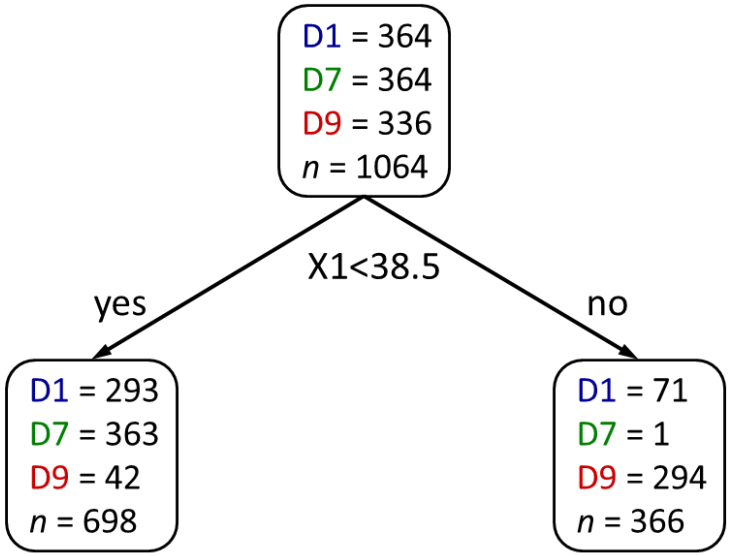
Recursive Partitioning

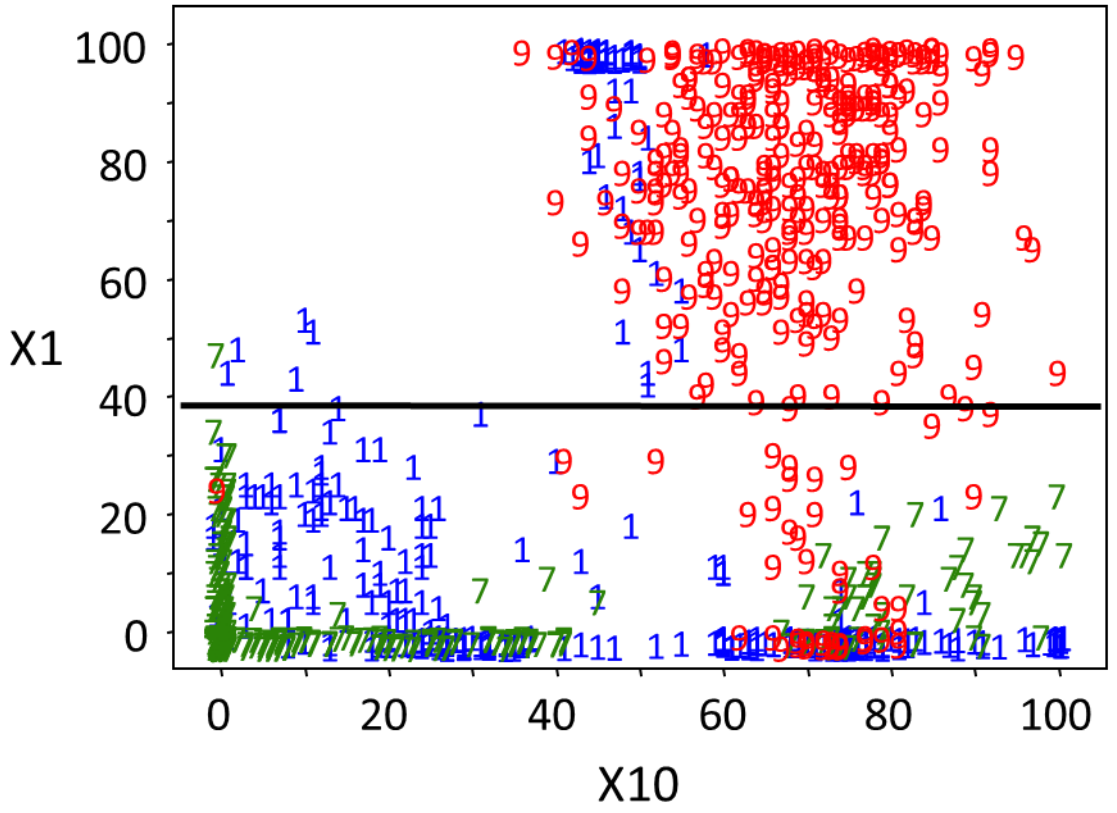
- Recursive partitioning is the standard method that is used to fit decision trees.
- It is a **top-down, greedy algorithm**.
- A greedy algorithm is an approach for solving a problem by selecting the best option available at the moment.
- Starting at the root node, several splits that involve a single input are examined.
- For interval inputs, the splits are disjoint ranges of the input values.
- For nominal inputs, the splits are disjoint subsets of the input categories.

Recursive Partitioning

- Various split-search strategies can be used to determine the set of candidate splits.
- A splitting criterion is used to choose the split.
- The goal of the splitting criterion is to reduce the variability of the target distribution in the child nodes.
- The cases in the root node are then partitioned according to the selected split.

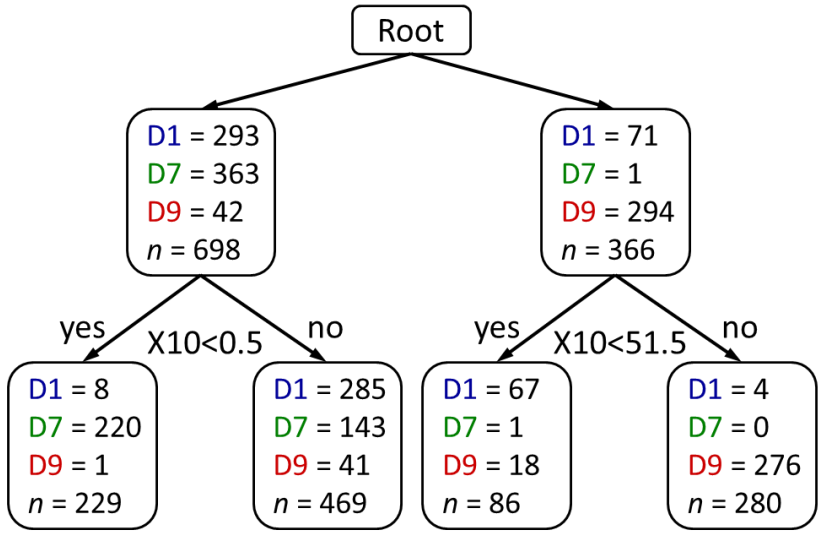
Recursive Partitioning

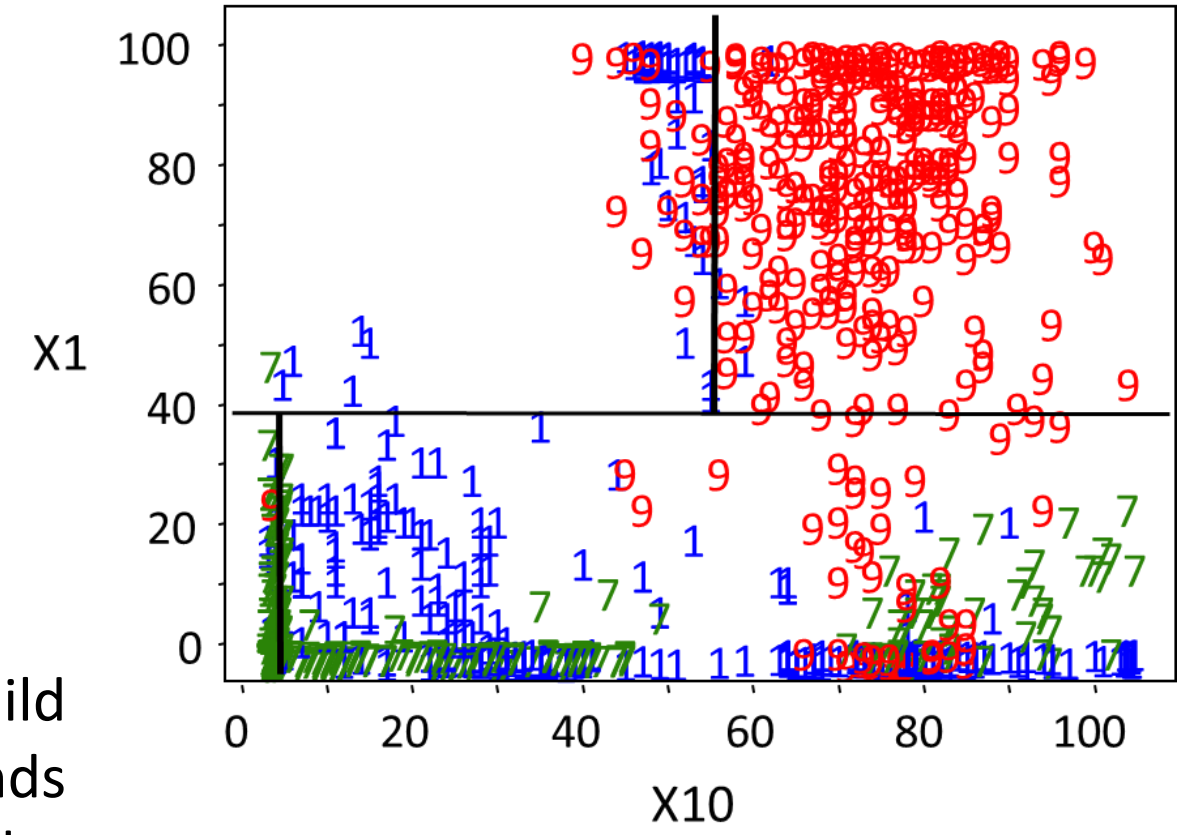




The root node split corresponds to a partition of the input where it gives the most significant reduction in variability, and the partition is perpendicular to that input dimension.

Recursive Partitioning





The partitioning is repeated in each child node. The split selection at node depends entirely on the cases in that local region of the input space.

Split Search

Partitioning on an Ordinal Input

Branches	Splits		
2	1—234 12—34 123—4	$\binom{3}{1} = 3$	Number of splits for a given number of branches
3	1—2—34 1—23—4 12—3—4	$\binom{3}{2} = 3$	$\binom{L-1}{B-1} = \frac{(L-1)!}{(B-1)!\,(L-B)!}$
4	1—2—3—4	$\binom{3}{3} = 1$	Total number of potential splits $\sum_{l=2}^{L}\binom{L-1}{l-1} = 2^{L-1}-1$

- Splits on ordinal inputs are restricted to preserve the ordering.
- Only adjacent values are grouped

Partitioning on an Ordinal Input

X	.20	1.7	3.3	3.5	14	2515
ln(X)	–1.6	.53	1.2	1.3	2.6	7.8
rank(X)	1	2	3	4	5	6



- For interval and ordinal inputs, splits in a decision tree depend only on the ordering of the levels.
- Therefore, decision tree models robust to outliers in input space.
- The application of a rank or any monotonic transformation to a numeric variable does not change the fitted tree.
- Monotonic transform invariant might not always hold when the splitting criteria depends on *p*-values

Partitioning on an Nominal Input

1—234
2—134
3—124
4—123
12—34
13—24
14—23

7 2-way splits

1—2—34
1—3—24
1—4—23
2—3—14
2—4—13
3—4—12

6 3-way splits

1—2—3—4

1 4-way split

$$S(L,B) = B \cdot S(L-1,B) + S(L-1,B-1)$$

L \ B:	2	3	4	total
2	1			1
3	3	1		4
4	7	6	1	14
5	15	25	10	51
6	31	90	65	202
7	63	301	350	876
8	127	966	1701	4139
9	255	3025	7770	21146

- Splits on a nominal input are not restricted.
- For a nominal input with L distinct levels, there are $S(L,B)$ partitions into B branches, where $S(L,B)$ is a Stirling number of the second kind.

$$S(L,B) = \frac{1}{B!}\sum_{j=0}^{B}(-1)^j \binom{B}{j}(B-j)^2$$

Split Search Shorcuts

- An exhaustive tree algorithm considers all possible partitions of all inputs at every node in the tree.
- The combinatorial explosion usually makes an exhaustive search expensive.
- Decision tree algorithms usually takes some shortcuts to reduce the split search. The shortcuts are:
  - Restricting searches to binary splits
  - Using level clustering routines
  - Imposing minimum child size restrictions

Splitting Criteria

Splitting Criteria

- After a set of candidate splits is determined, a splitting criterion is used to determine the best one.
- For classification trees, the three most well-known splitting criteria are based on the Gini index (BFOS 1984), entropy (Quinlan 1993), and the chi-square test (Kass 1980), the related algorithms are CART, C5.0 and CHAID, respectively.

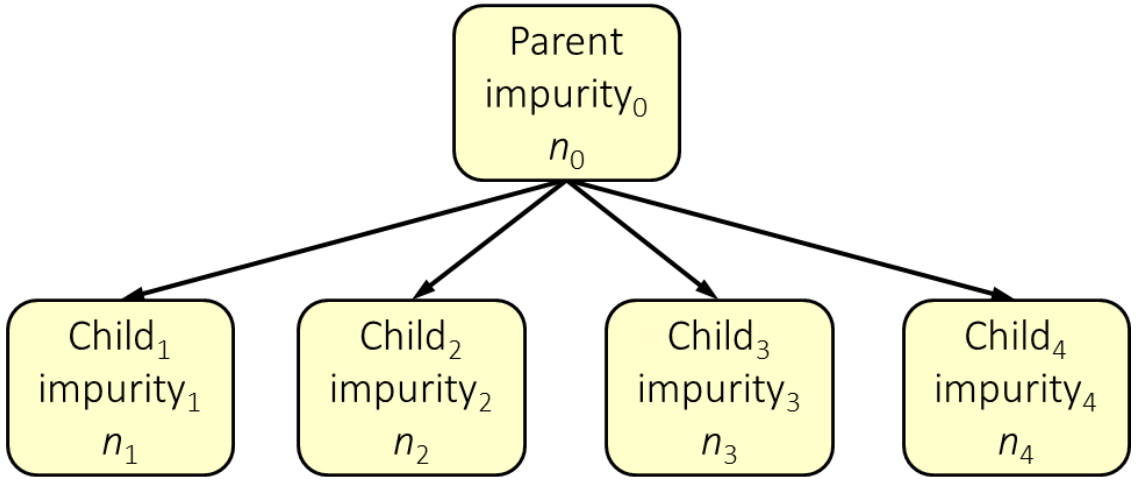
X1:	<38.5	≥38.5	ΔGini	Δentropy	Logworth	*
1	293	71				
7	363	1	.197	.504	140	
9	42	294				

X10:	<0.5	1-41	42-51	≥51.5			
1	9	143	65	147			
7	221	88	1	54	.255	.600	172
9	1	4	16	315			

* ΔGini and Δentropy are impurity reduction measures. Logworth is based on the χ^2 from an *n-way* table. For all, higher is better.

Impurity Reduction Measures

- $i(.)$ is some measure of within-node impurity and Δi represents the overall reduction in impurity for the tree.
- Many splitting criteria are based on the reduction in node impurity induced by the split.



$$\Delta i = i(0) - \left(\frac{n_1}{n_0} i(1) + \frac{n_2}{n_0} i(2) + \frac{n_3}{n_0} i(3) + \frac{n_4}{n_0} i(4) \right)$$

Gini Impurity

- The Gini index is a measure of variability for categorical data.
- A pure node has a Gini index of 0.
- As the number of evenly distributed classes increases, the Gini index approaches 1.
- So ΔGini is higher better.

$$1 - \sum_{j=1}^{r} p_j^2 = 2 \sum_{j<k} p_j p_k$$



Pr(interspecific encounter) = 1-2(3/8)^2-2(1/8)^2 = .69

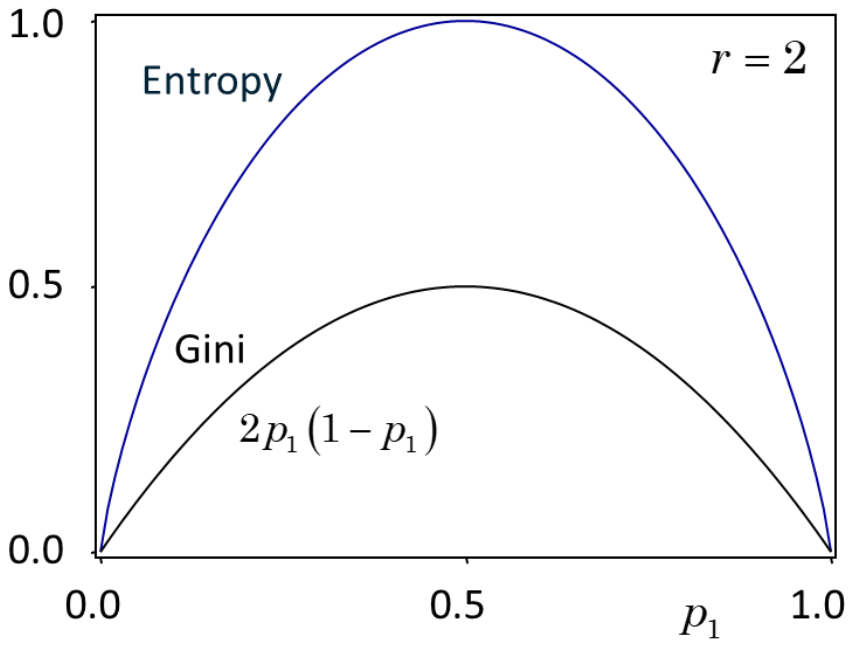


Pr(interspecific encounter) = 1-(6/7)^2-(1/7)^2 = .24

Relationship of Entropy and Gini Functions

- Entropy is also a measure of variability for categorical data.
- Entropy is the average rarity and thus measures the uncertainty of the outcome.
- Δ entropy splitting criterion is equivalent to using the likelihood ratio chi-square test statistic for association between the branches and the target categories. (Ripley 1996)

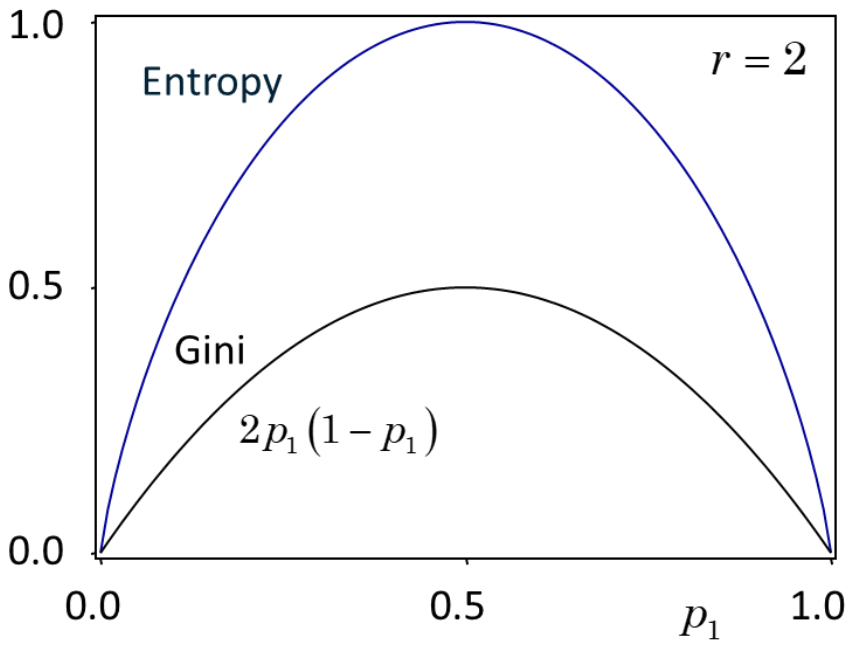
$$\text{Entropy}=H\left(p_1,p_2,\ldots,p_r\right)=-\sum_{i=1}^{r}p_i\log_2\left(p_i\right)$$



Relationship of Entropy and Gini Functions

- Quinlan(1993) prefers the use of gain ratio, a modification of Δ entropy that reduces its bias toward a larger number of branches and attributes with many values.
- Breiman (1996) showed that the Δ Gini criterion tends to favor isolating the largest target class in one branch, whereas the Δentropy criterion tends to favors split balance.

$$\text{Entropy}=H\left(p_1,p_2,\ldots,p_r\right)=-\sum_{i=1}^{r}p_i\log_2\left(p_i\right)$$



Chi-Square Test

- The split in a classification tree can be arranged in a contingency table .
- The rows or columns represent either the child nodes or the target classes.
- Each cell contains the frequency of cases of that class in that child node.

Observed

X1:	<38.5	≥38.5	
1	293	71	.342
7	363	1	.342
9	42	294	.316
	.656	.344	*n*=1064

Expected

239	125
239	125
225	116

$\frac{(O-E)^2}{E}$

12	23
64	123
149	273

$$\text{Expected} = \textit{row proportion} \times \textit{column proportion} \times n$$

$$\chi^2_\nu = \sum \frac{(O-E)^2}{E} = 644 \qquad \nu = (3-1)(2-1) = 2$$

Chi-Square Test

- The *p*-value of the χ^2 is the probability that how likely the columns and rows are no association.
- The chi-square splitting criterion uses the *p*-value of the chi-square test
- When the *p*-values are very small, it is more convenient to use $\text{Logworth} = -\log_{10}(p\text{—value})$

Observed

X1:	<38.5	≥38.5	
1	293	71	.342
7	363	1	.342
9	42	294	.316
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$$\chi^2_\nu = \sum \frac{(O-E)^2}{E} = 644 \qquad \nu = (3-1)(2-1) = 2$$

Chi-Square Test

- The ΔGini and Δentropy splitting criteria also tend to increase as the number of branches increase.
- They do not have an degree of freedom adjustment like *p*-value of Chi-square test statistic.
- Consequently, they favour multi-way split with large *B*.

Observed

X1:	<38.5	≥38.5	
1	293	71	.342
7	363	1	.342
9	42	294	.316
	.656	.344	*n*=1064

Expected

239	125
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225	116

$\frac{(O-E)^2}{E}$

12	23
64	123
149	273

$$\text{Expected} = \textit{row proportion} \times \textit{column proportion} \times n$$

$$\chi^2_\nu = \sum \frac{(O-E)^2}{E} = 644 \qquad \nu = (3-1)(2-1) = 2$$

Chi-Square Test

- Split search needs adjustments to control the possible spurious associations discovered because too many splits were examined.
- Kass (1980) proposed Bonferroni adjustments of the *p*-values to account for this bias.
- For a set of *m* tests, the Kass adjustment multiplies the *p*-values by *m*.
- The multiplier *m* is $\binom{L-1}{B-1}$ for ordinal inputs and $S(L,B)$ for nominal inputs

X1: 38.5

1	293	71
7	363	1
9	42	294

X1: 17.5 36.5

1	249	42	73
7	338	25	1
9	26	16	294

X10: 0.5 41.5 51.5

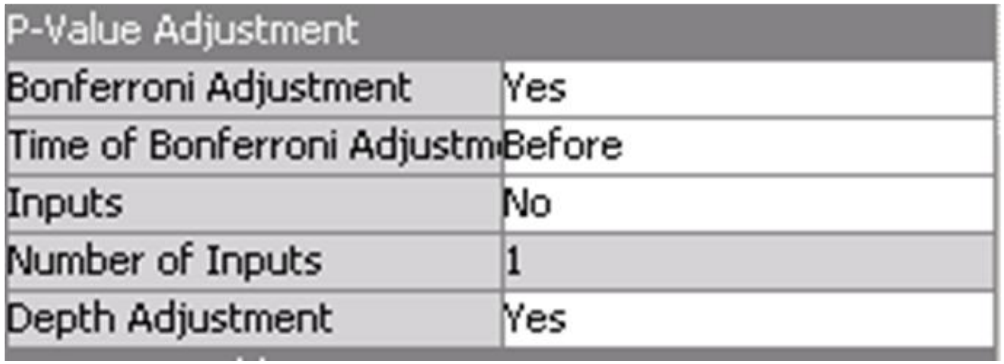
1	9	143	65	147
7	221	88	1	54
9	1	4	16	315

χ^2_ν	ν	$-\log_{10}(\mathrm{P})$	m	$-\log_{10}(m\mathrm{P})$
644	2	140	96	138
660	4	141	4560	137
814	6	172	156849	167

Splitting Criterion and *p*-value Adjustment

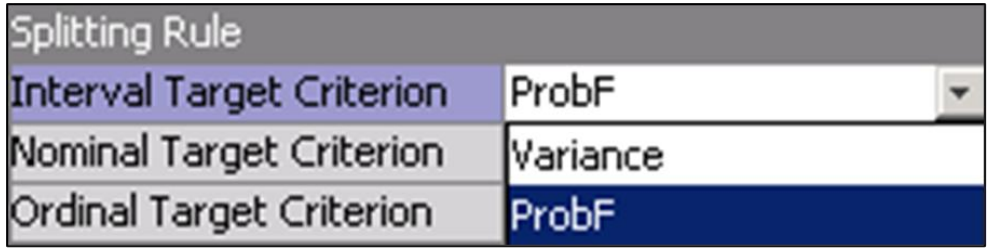
Splitting Criterion and *p*-value Adjustment on Decision Tree Node property panel.





Splitting Criterion for Interval Targets

- Regression trees try to partition the input space into segments where the variance of the target is zero.
- The split-search considerations and *p*-value adjustments are the same as with classification trees.
- But the splitting criteria are different.



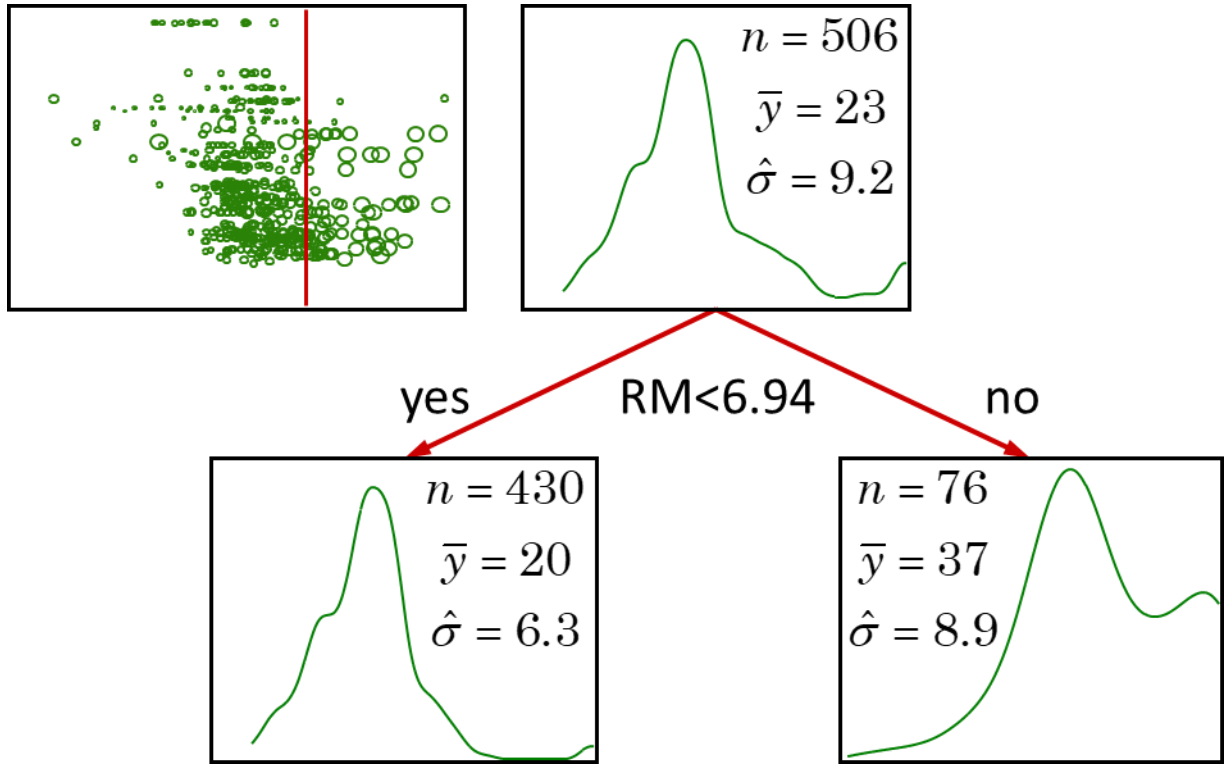
The default splitting criterion is ProbF (The p-value of the F test associated with an ANOVA comparing the means of the child nodes in a split.

Splitting Criterion for Interval Targets

- The sample variance is the obvious measure of impurity for continuous target.

$$i(t) = \frac{1}{n_t} \sum_{j=1}^{n_t} \left(y_{jt} - \bar{y}_t \right)^2$$

- The denominator is n_i not $n_i - 1$. This is the MLE of sample variance.



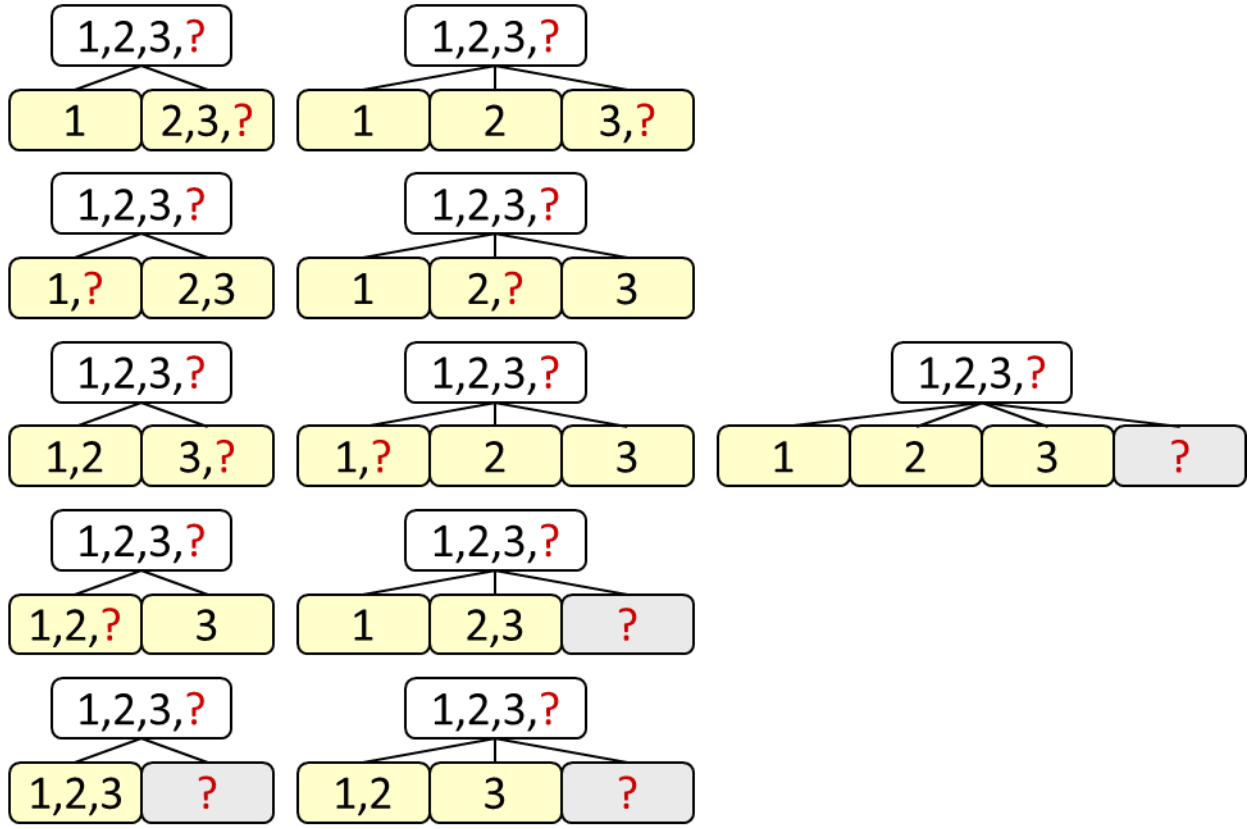
Missing Values

Handling Missing Values in Decision Trees

- Nominal inputs: Treat missing values as a separate level.
- Ordinal inputs: Require modification of the split search strategy for missing values
- Additional options:
  - Largest branch: Assign the cases that contain missing values to the largest branch.
  - Most correlated branch: Assign the cases to the branch with the smallest residual sum of squares among that contain missing values.

Handling Ordinal Missing Values in Decision Trees

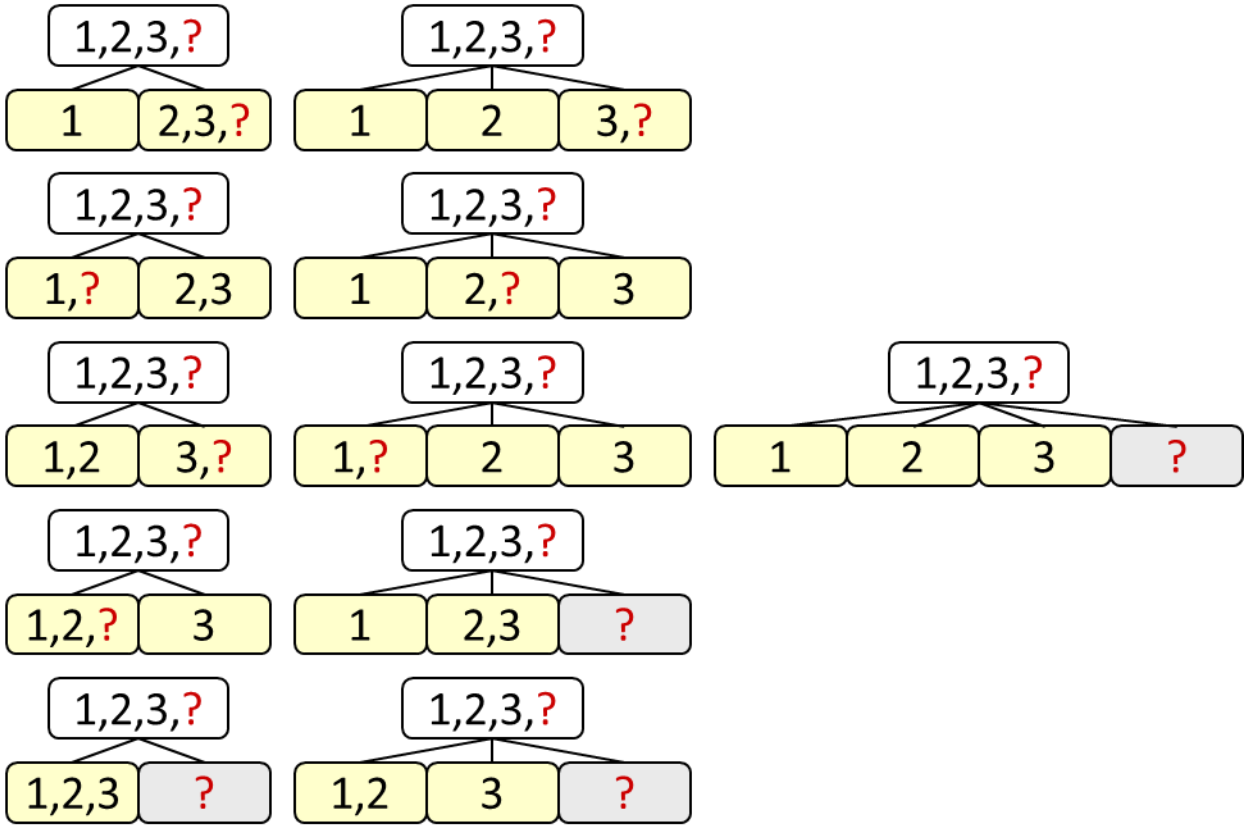
- An ordinal input with missing values requires modifications of the split search strategy and the Bonferroni multipliers
- The missing values cannot be placed in order among the input level.
- The missing value acts as a nominal level.
- So, split search cannot restrict the branch with missing level.



Handling Ordinal Missing Values in Decision Trees

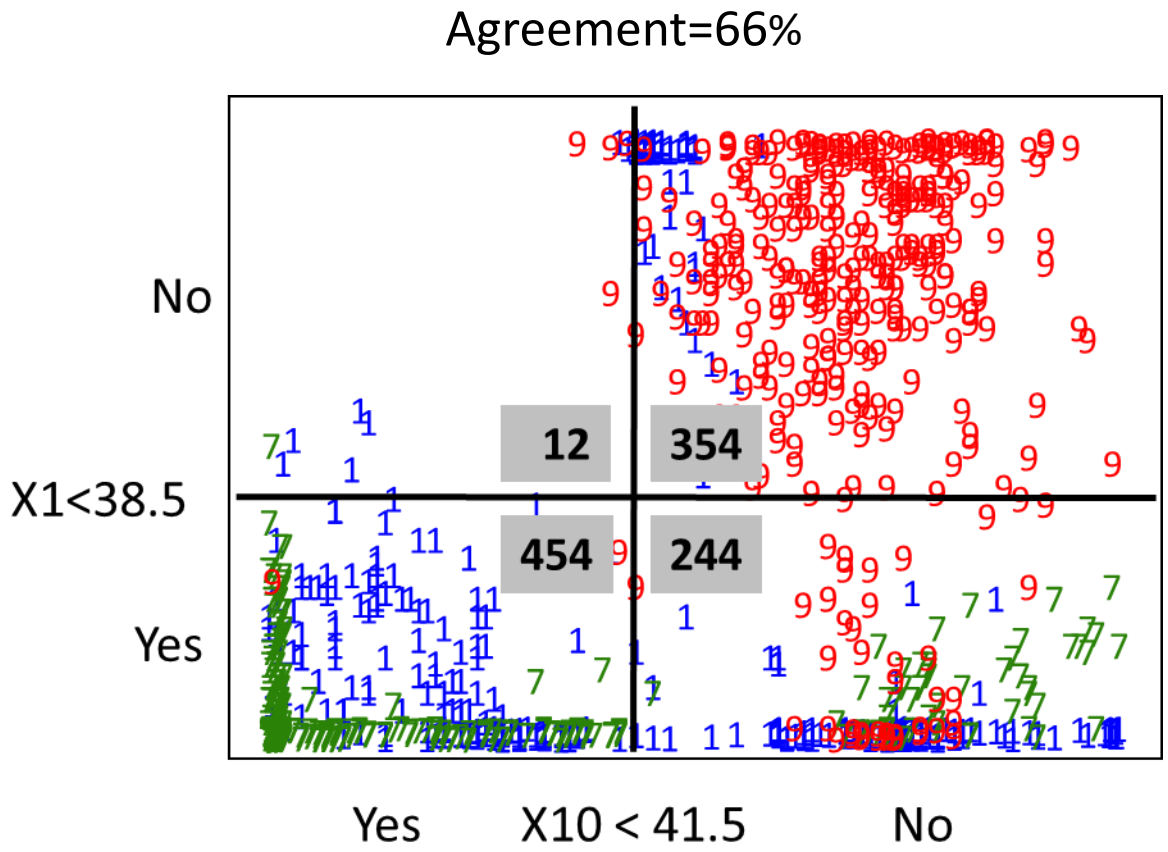
- The branch with missing level might be in a branch by itself or mixed in with the adjacent ordinal levels
- This increases the number of splits to consider from

$$\binom{L-1}{B-1} \text{ to } B\binom{L-1}{B-1} + \binom{L-1}{B-2}$$



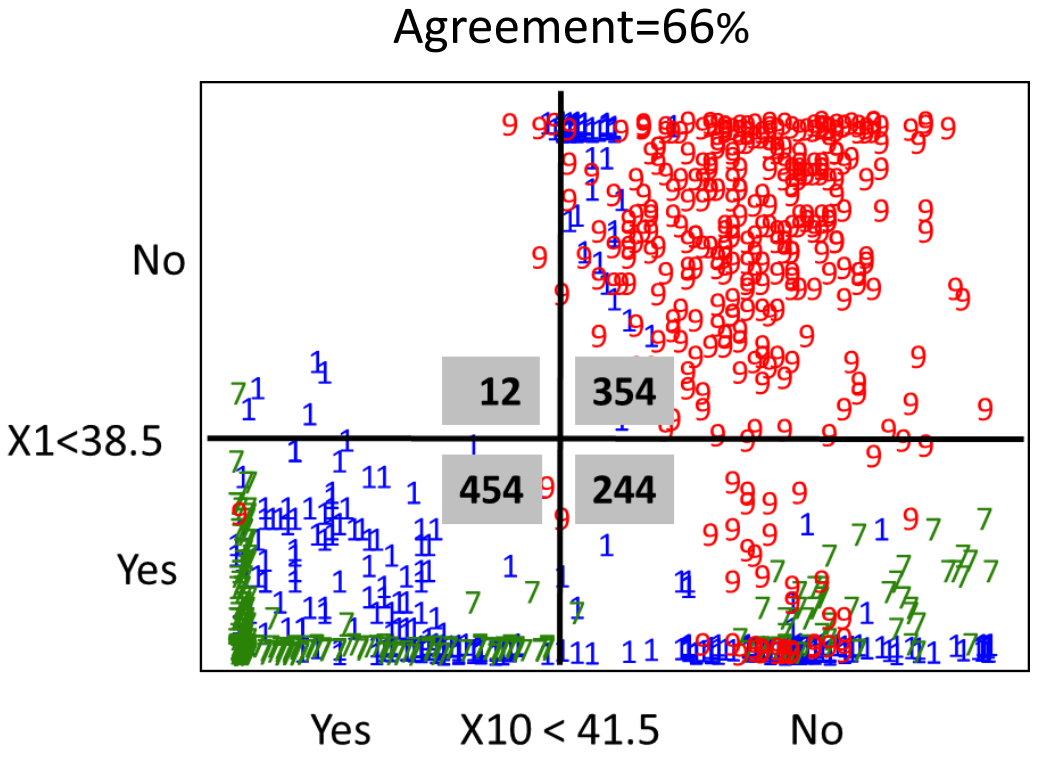
Surrogate Splits

- Surrogate splits can be used to handle missing values.
- A surrogate split is a partition using a different input that mimics the selected split.
- The agreement between two splits can be measured as the proportion of cases that are sent to the same branch.
- The split with the greatest agreement is taken as the best surrogate.



Surrogate Splits

- Agreement measures the proportion of observations that have a nonmissing value in the rule's predictor variable that two rules assign to the same branch.
- For example, if the primary splitting rule assigns 100 observations to a child node, surrogate rule A assigns 40 observations to the same child node, and surrogate rule B assigns 50 observations to the same child node, then surrogate rule B is applied before surrogate rule A.

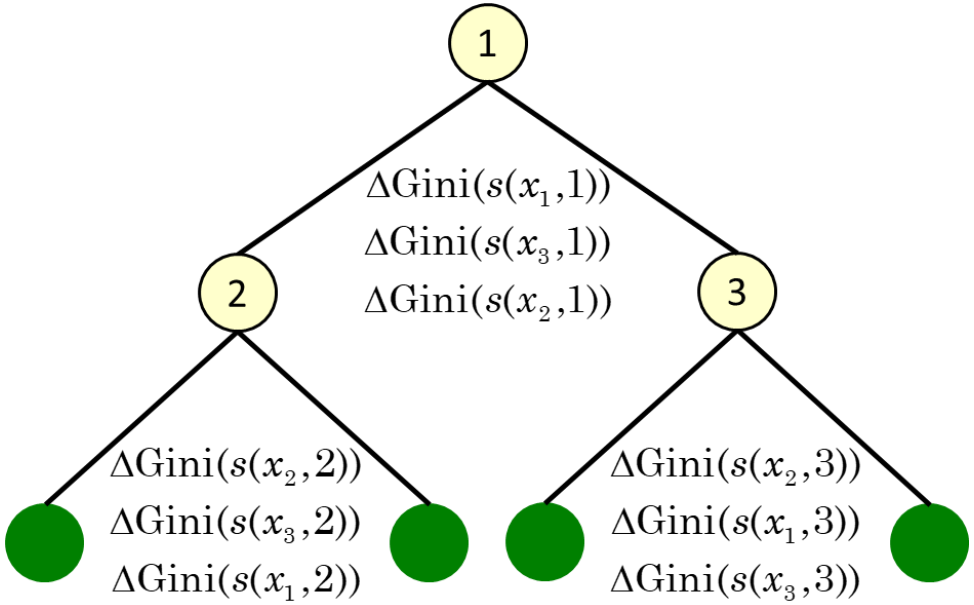


Variable Importance

Variable Importance

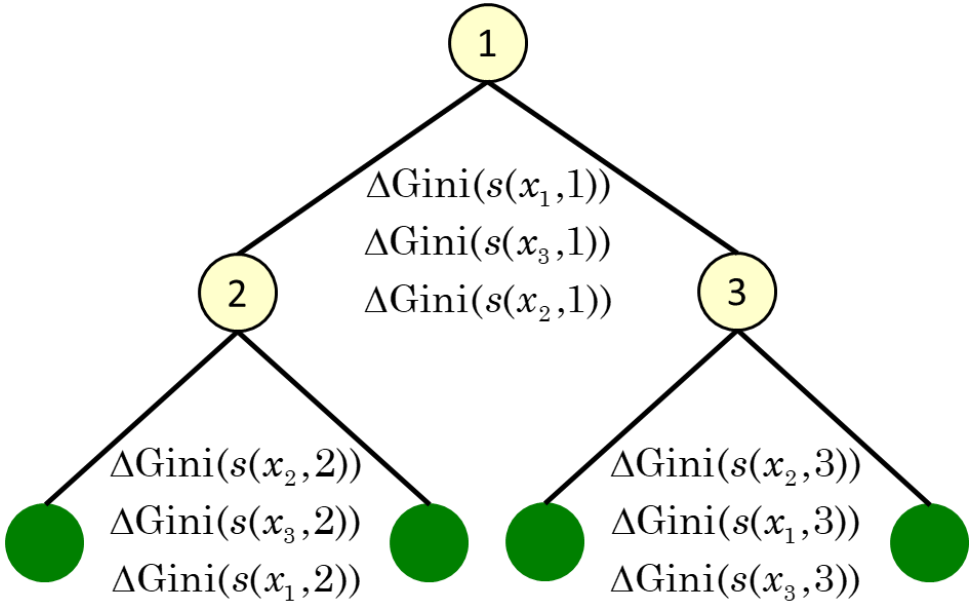
- BFOS (1984) devised a measure of variable importance for decision trees.
- Let $s(x_j, t)$ be a surrogate split (including the primary split) at the tth internal node using the jth input.
- Importance is a weighted average of the reduction in impurity for the surrogate splits using the jth input across all the internal nodes in the decision tree.
- The weights are the node size

$$\text{Importance}(x_j) = \sum_{i=1}^{T} \frac{n_t}{n} \Delta(i) s(x_j, t)$$



Variable Importance

- In the **Decision Tree node**, variable importance is calculated according to BFOS (1984), except by taking the square root.
- Further, the calculation incorporates the agreement between the surrogate split and the primary split.
- And the variable importance measure is scaled to be between 0 and 1 by dividing by the maximum importance.
- Variables that do not appear in any primary or saved surrogate splits have 0 importance.



Observation-Based Variable Importance

- The observation-based approach involves reading a data set and applying the model several times to each observation.
- First, the standard prediction is computed.
- Then each variable is made uninformative, and a prediction is computed.
- Each variable is made uninformative by assigning an observation to both sides of a split.
- The prediction of observation is the weighted average of predictions over the leaves into which the observation is assigned.

Increasing Importance

X	ΔASE	ASE*x*	ASE*o*
LSTAT	0.1242	0.1488	0.0246
CRIM	0.0228	0.0474	0.0246
NOX	0.0172	0.0418	0.0246
DIS	0.0027	0.0273	0.0246
TAX	0.0019	0.0265	0.0246

$$\Delta x = fx - fo$$

Observation-Based Variable Importance

$$\Delta x = fx - fo$$

Where *fx* denotes a specific fit statistic value when variable *x* is made uninformative, and *fo* is the value when *x* is not made uninformative.

- The observation-based variable importance is misleading when some variables are correlated.

Increasing Importance

X	ΔASE	ASE*x*	ASE*o*
LSTAT	0.1242	0.1488	0.0246
CRIM	0.0228	0.0474	0.0246
NOX	0.0172	0.0418	0.0246
DIS	0.0027	0.0273	0.0246
TAX	0.0019	0.0265	0.0246

Lesson Summary

- Recursive partitioning is a top-down, greedy algorithm based on split search strategies with splitting criteria for useful inputs selection in decision trees.
- The principle of split search strategies is to reduce the impurity or variability in the child nodes.
- The Gini index, entropy and the Chi-square test are the splitting criteria for categorical targets. And ProfF and Variance reduction are the splitting criteria for continuous targets.
- Nominal missing values are treated as an individual level. Ordinal missing values are placed in a single branch or mixed with the adjacent ordinal levels. Continuous missing values are handled by using surrogate splits.
- Variable importance is used to measure the significance of an input in a decision tree.

How does a decision tree optimize its complexity?



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Topic 5.3
Supervised Learning: Tree-Based Models

Optimising Complexity: Pruning

By Dr Lee How Chinh



Objectives

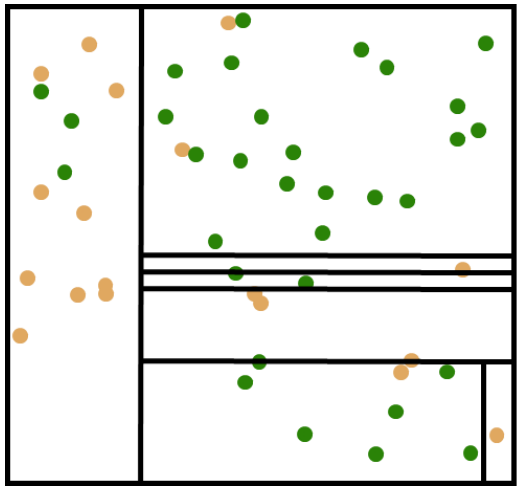
- Describe decision trees model complexity and generalisation
- Compare top-down and bottom-up pruning methods
- Explain pruning criteria and sub-tree selection

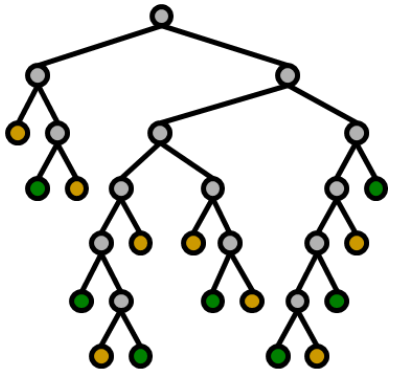
Contents

- Decision Trees Complexity And Generalisation
- Top-Down Stopping Rules
- Bottom-Up Pruning
- Model Selection Criterion

Decision Trees Complexity And Generalisation

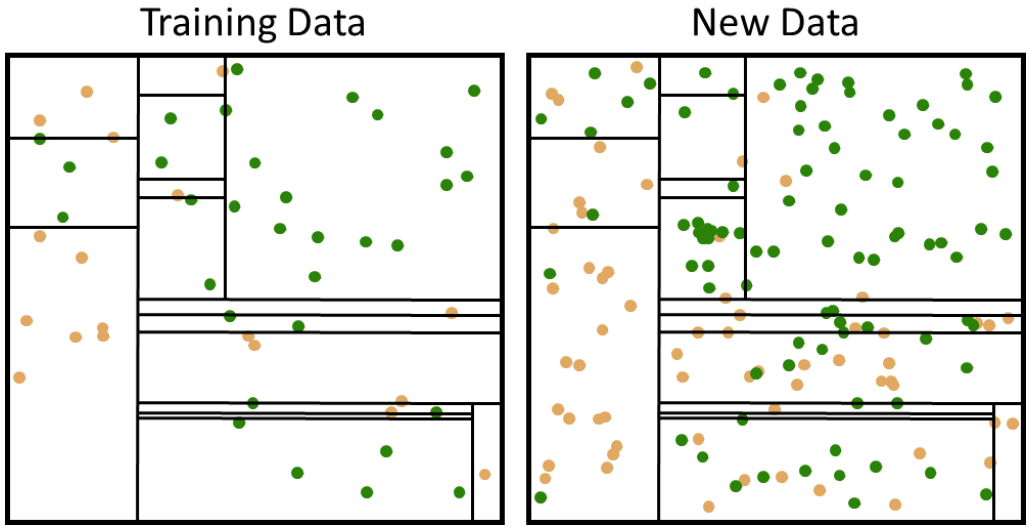
Maximal Tree





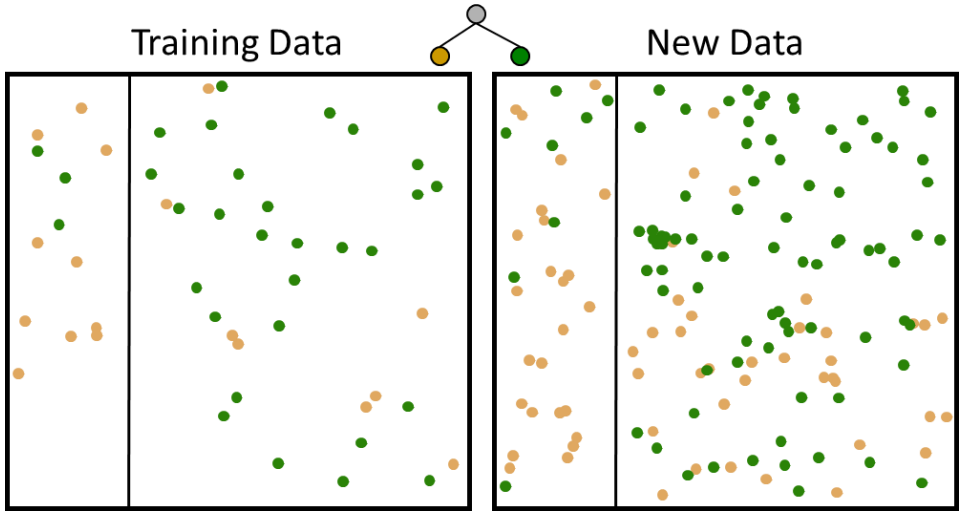
- The decision tree with the greatest possible purity on the training data is the **maximal classification tree**.

Overfitting



- The maximal tree is an overfitting tree.
- It adapts to both the systematic variation of the target (signal) and the random variation (noise).
- It usually does not generalise well on new data.

Underfitting



- A small tree with only a few branches might underfit the data.
- It can fail to adapt sufficiently to the signal.
- It is usually poor in generalisation.
- Decision tree complexity is a function of the number of leaves, the number of splits, and the depth of the tree.

Pruning for the right-sized tree

- Pruning refers to the various methods for selecting decision tree complexity.
- There are two pruning methods:
  - Top-Down Stopping Rules (Pre-Pruning)
  - Bottom-Up Selection Criteria (Post-Pruning)

Top-Down Stopping Rules

Top-Down Stopping Rules

Top-Down Stopping Rules can be based on the following conditions:

- limiting the depth of the tree
- limiting the amount of fragmentation
- statistical significance

Top-Down Stopping Rules

Limiting the depth of the tree

- A decision tree stops splitting if the tree achieves the threshold.

Limiting the amount of fragmentation

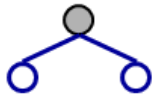
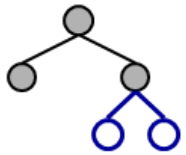
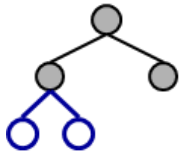
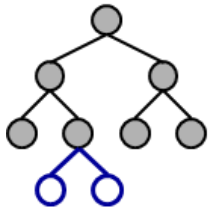
- A decision tree stops splitting a node if the number of cases drops below the threshold.

Top-Down Stopping Rules

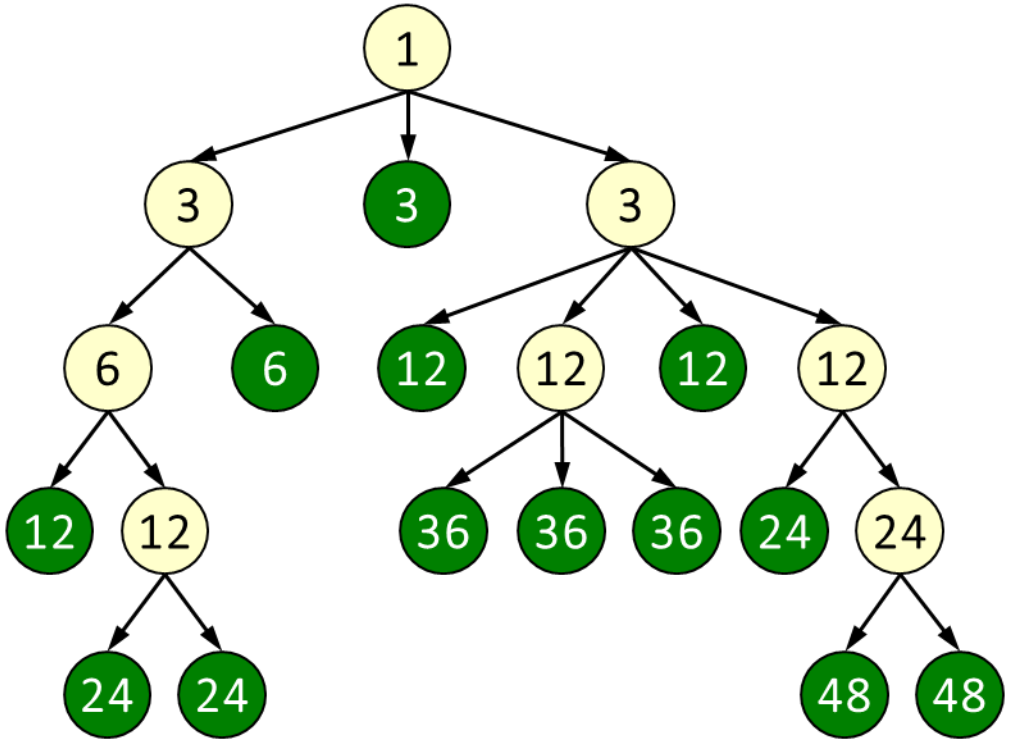
Statistical significance

- The decision tree node has an adjustment that depends on the depth of the split.
- For a binary tree, the multiplier is 2^d where d is the depth of the split

Top-Down Stopping Rules

	$-\log_{10}(\mathrm{P})$	m	$-\log_{10}(m\mathrm{P})$	d	$-\log_{10}(2^d m\mathrm{P})$
	26.7	53	24.9	0	24.9
	3.12	14	1.97	1	1.67
	1.63	39	.039	1	−.26
	2.40	11	1.36	2	.76

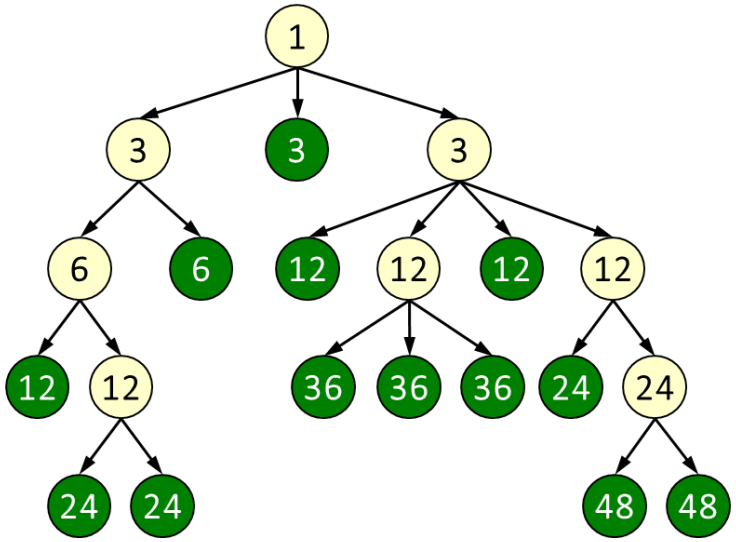
Depth Multiplier



Depth Multiplier=$\prod_{i=0}^{d} B_i$, where B_i is the number of branches

Depth Multiplier

- The depth adjustment controls the familywise Type I error rate for the tests at all nodes.
- Let the desired familywise Type I error rate is α .
- If each individual test in the family is rejected only if the *p*-value is smaller than $\frac{\alpha}{(\text{depth multiplier})}$, then the familywise Type I error rate is less than or equal to α.
- For this to be true, the sum over all the leaves of the reciprocals of the multipliers must equal to one.



Decision Tree Node Options for Top-Down Pruning

- **Significance Level:** The default is 0.2 (logworth = 0.7)
- **Maximum Depth:** The default is 6.
- **Leaf Size:** Specify the smallest number of leaves must consider for splitting. Valid values are between 2 to 32767.
- **Subtree Method: Largest**
  For Top-down pruning
- **Depth Adjustment:** By default, the depth multiplier is applied.

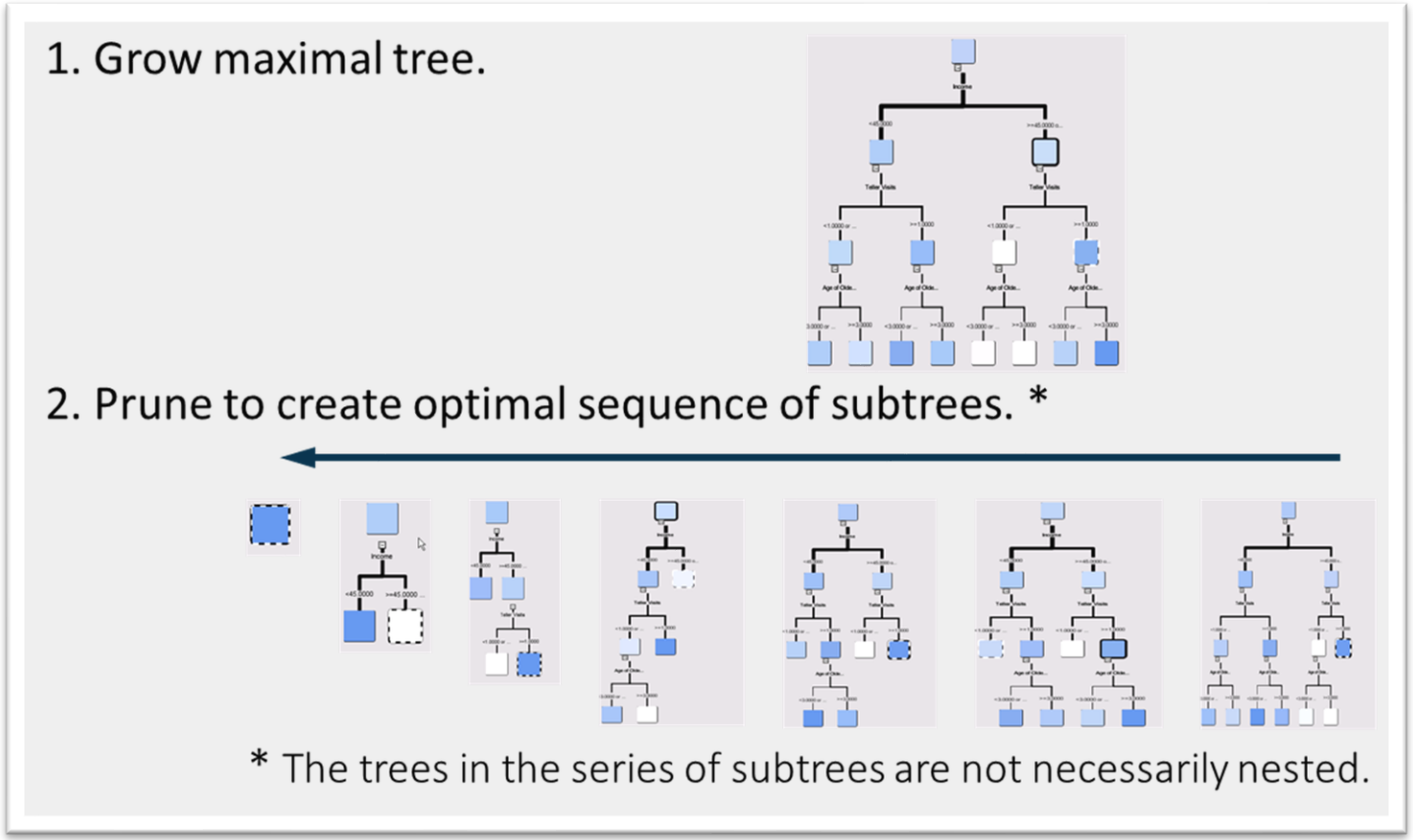
Splitting Rule	
Interval Target Criterion	ProbF
Nominal Target Criterion	Gini
Ordinal Target Criterion	Entropy
Significance Level	0.2
Missing Values	Use in search
Use Input Once	No
Maximum Branch	2
Maximum Depth	10
Minimum Categorical Size	5
Node	
Leaf Size	60
Number of Rules	5
Number of Surrogate Rules	2
Split Size	.
Split Search	
Subtree	
Method	Largest
Number of Leaves	Assessment
Assessment Measure	Largest
Assessment Fraction	N

P-Value Adjustment	
Bonferroni Adjustment	Yes
Time of Bonferroni Adjustm	Before
Inputs	No
Number of Inputs	1
Depth Adjustment	Yes

Bottom-Up Pruning

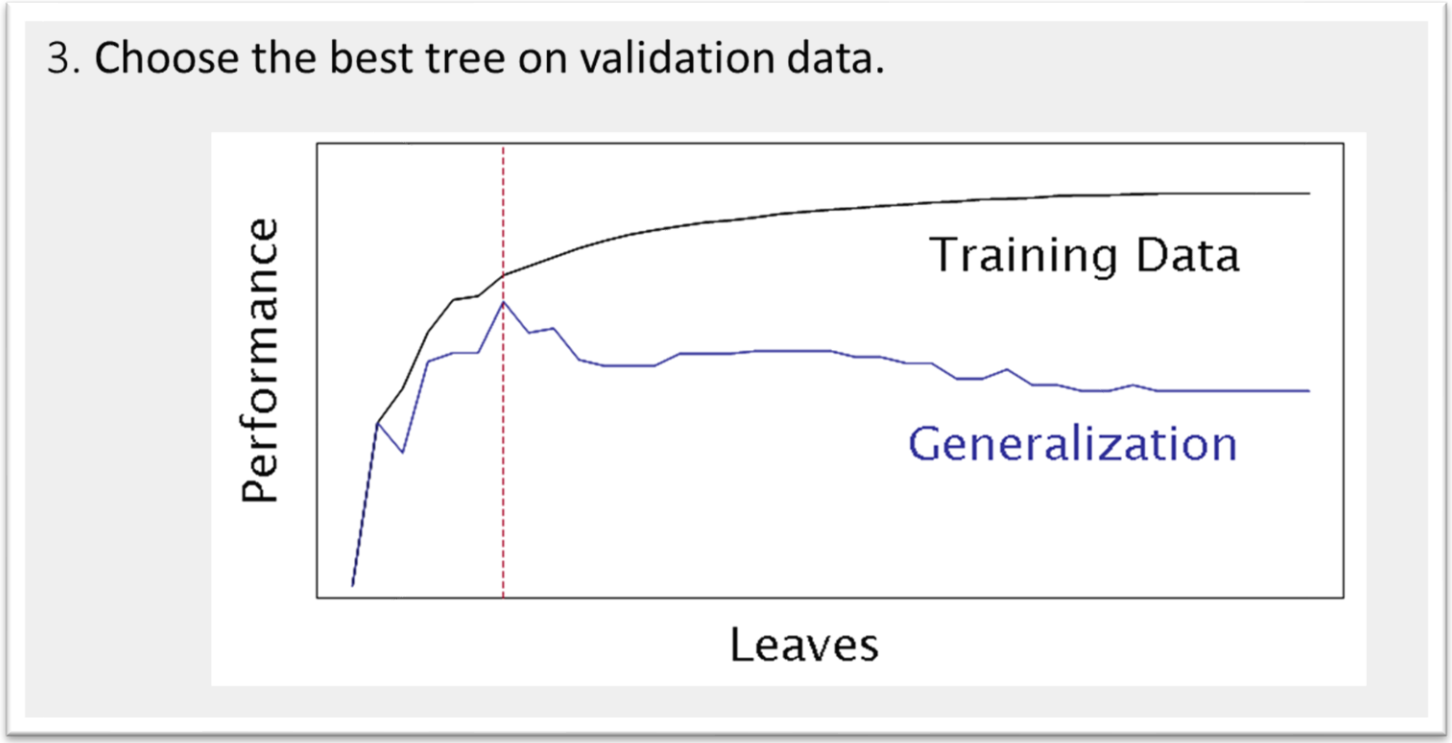
Bottom-Up Pruning

- A large tree is grown then branches are removed in a backward fashion using some model selection criterion.
- It is also called retrospective pruning.



Bottom-Up Pruning

- The subtree with the best performance on validation data is selected.

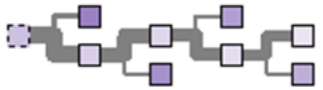
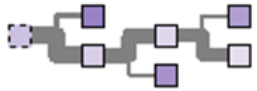
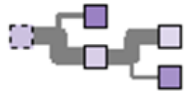
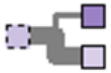


Bottom-Up Pruning

- Top-down pruning is usually faster but is considered less effective than bottom-up pruning
- Bottom-up pruning has two requirements:
  - A method for honestly measuring performance
    - Data partition (training and validation data sets)
  - A relevant selection criterion

Model Selection Criterion

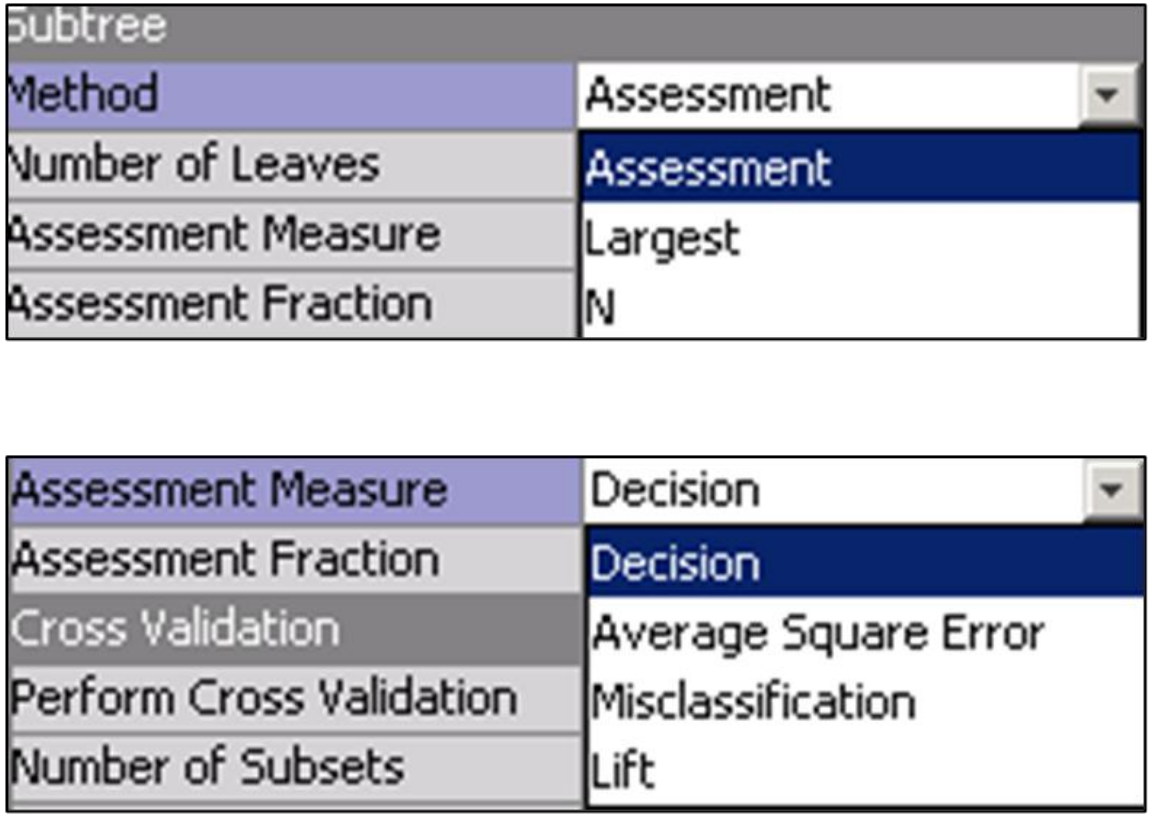
Model Selection Criteria

Leaves	Accuracy	Profit	ASE	
5	.90/.88	.51/.43	.17/.15	
4	.90/.88	.51/.40	.18/.14	
3	.89/.91	.49/.44	.19/.16	
2	.88/.91	.49/.44	.20/.16	
1	.59/.64	.04/ −.1	.48/.46	

*The first number is for training data. The second is for validation data.

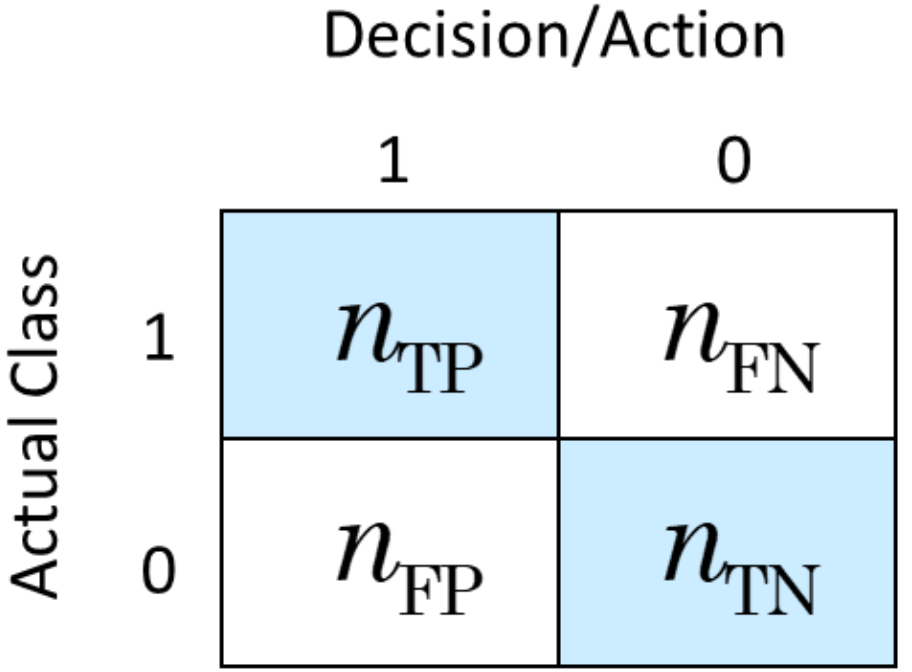
Decision Tree Node Options for Bottom-Up Pruning

- **Method: Assessment**. The final decision tree is selected based on the assessment measure.
- **Assessment Measure:** The default decision tree selection criterion is **Decision**. The selection defaults to **Average Square Error** or **Misclassification** for interval or categorical targets, respectively. **Lift** restricts assessment to a specified proportion of the data. The default percentage is 0.25.

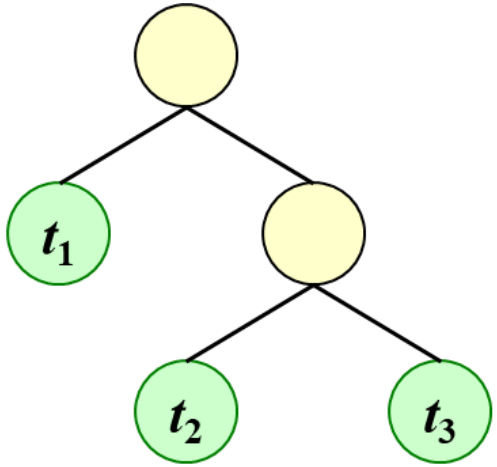


Confusion Matrix

- The confusion matrix is a cross tabulation of actual and predicted classes.
- A confusion matrix requires a decision rule.
- For binary target, the cutoff is 50% of the posterior probability.



Decision Tree Accuracy



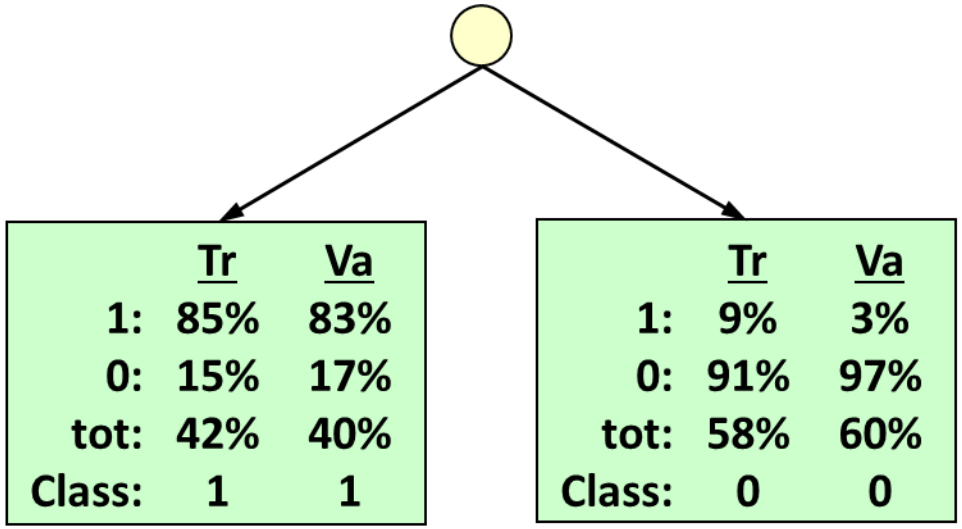
$$\text{Accuracy} = \frac{1}{n}\left(n_1(t_1)\,acc(t_1) + n_2(t_2)\,acc(t_2) + n_3(t_3)\,acc(t_3)\right)$$

- The accuracy of a tree can be computed as a weighted average of the accuracy in each leaf.
- The weights are the node sizes.

Decision Tree Accuracy

For training accuracy,

- 42% fall into left branch and all are classified as 1.
- Of those, 85% are classified correctly.
- So, the accuracy in the left branch would be $(0.42)(0.85)$



Training Accuracy = (0.42)(0.85) + (0.58)(0.91) = 0.88

Validation Accuracy = (0.40)(0.83) + (0.60)(0.97) = 0.91

Lesson Summary

- A maximal tree tends to be overfitting, and a small tree can be underfitting.
- Pruning is a method for obtaining an optimal decision tree
- There are two pruning methods, top-down stopping rules and Bottom-up selection criteria.
- The top-down stopping rule is a forward selection method, whereas, Bottom-up pruning is a backward elimination method.
- The top-down rule can be based on the depth of the tree, number of cases in nodes and statistical significance.
- Bottom-up pruning needs a method for honest assessment with a selection criterion.

Are there any advanced tree-structured models?



MONASH University



Topic 5.4
Supervised Learning: Tree-Based Models

Ensembles of Trees

By Dr Lee How Chinh



Objectives

- Describe decision tree instability
- Define the perturb and combine methods
- Describe bagging, boosting and gradient boosting algorithms.
- Describe forest models

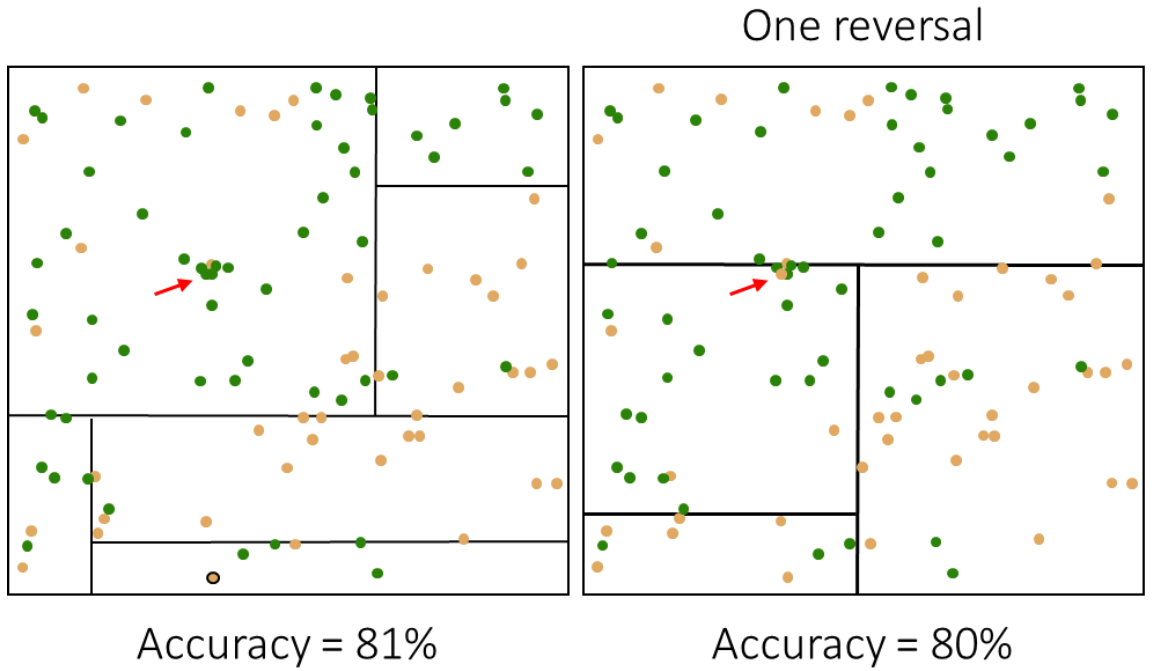
Contents

- Ensembles
- Bagging, Boosting and Gradient Boosting
- Forest Models

Ensembles

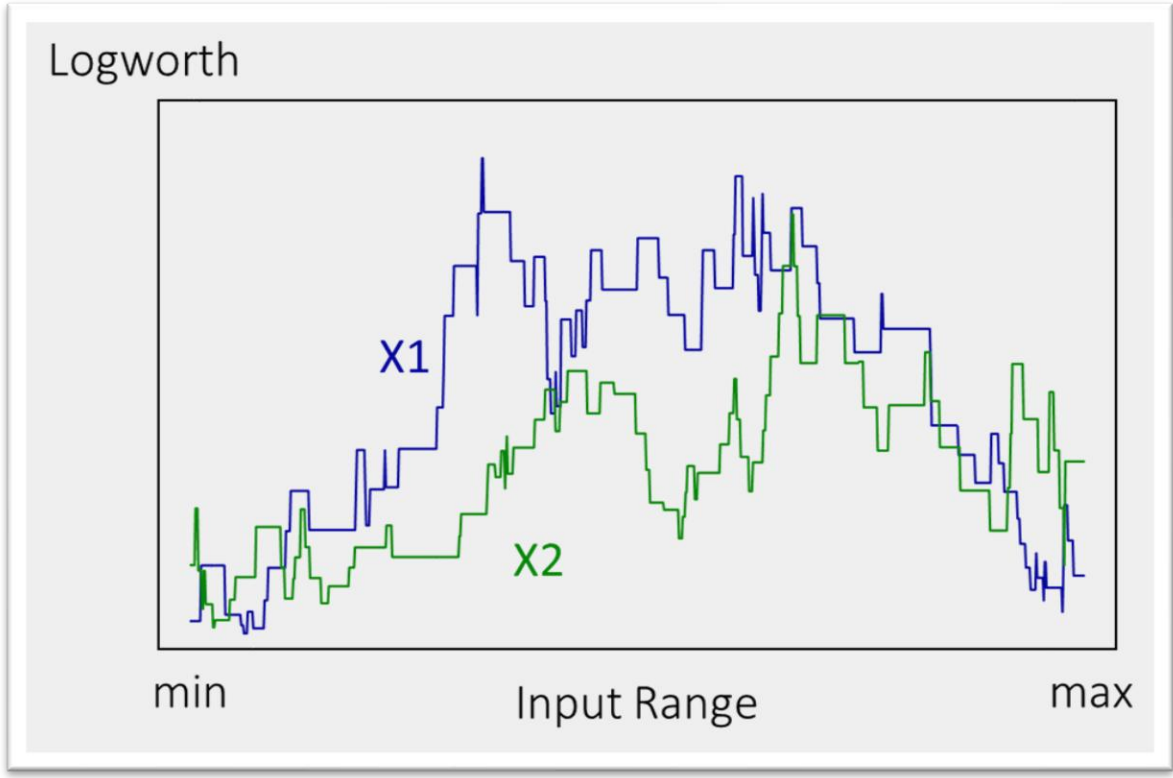
Instability

- Decision trees are unstable model.
- Small changes in the training data can cause obvious changes in the topology if the tree.
- But the overall performance of the decision tree remains stable.



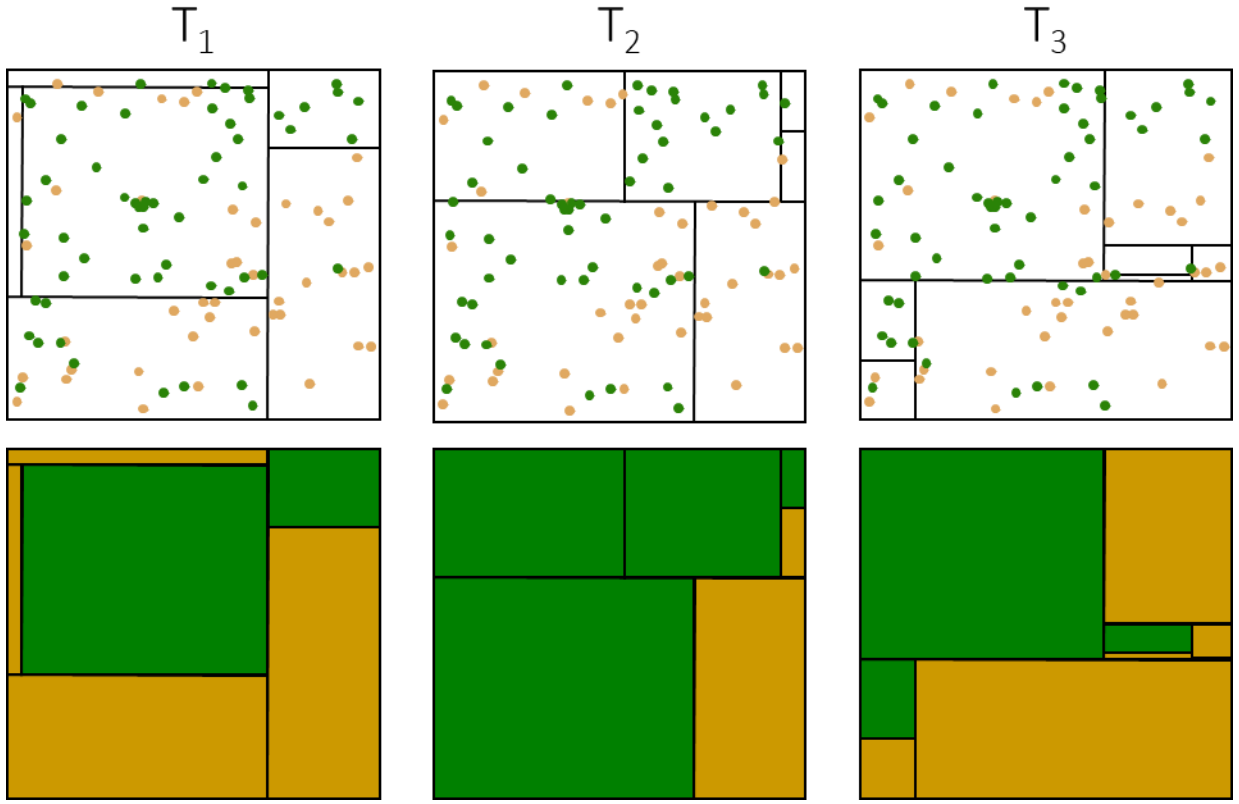
Instability

- The instability causes from the large number of univariate splits and the fragmentation of the data.
- There are many competitor splits at each split.
- So a small change in the data can easily result a different split.



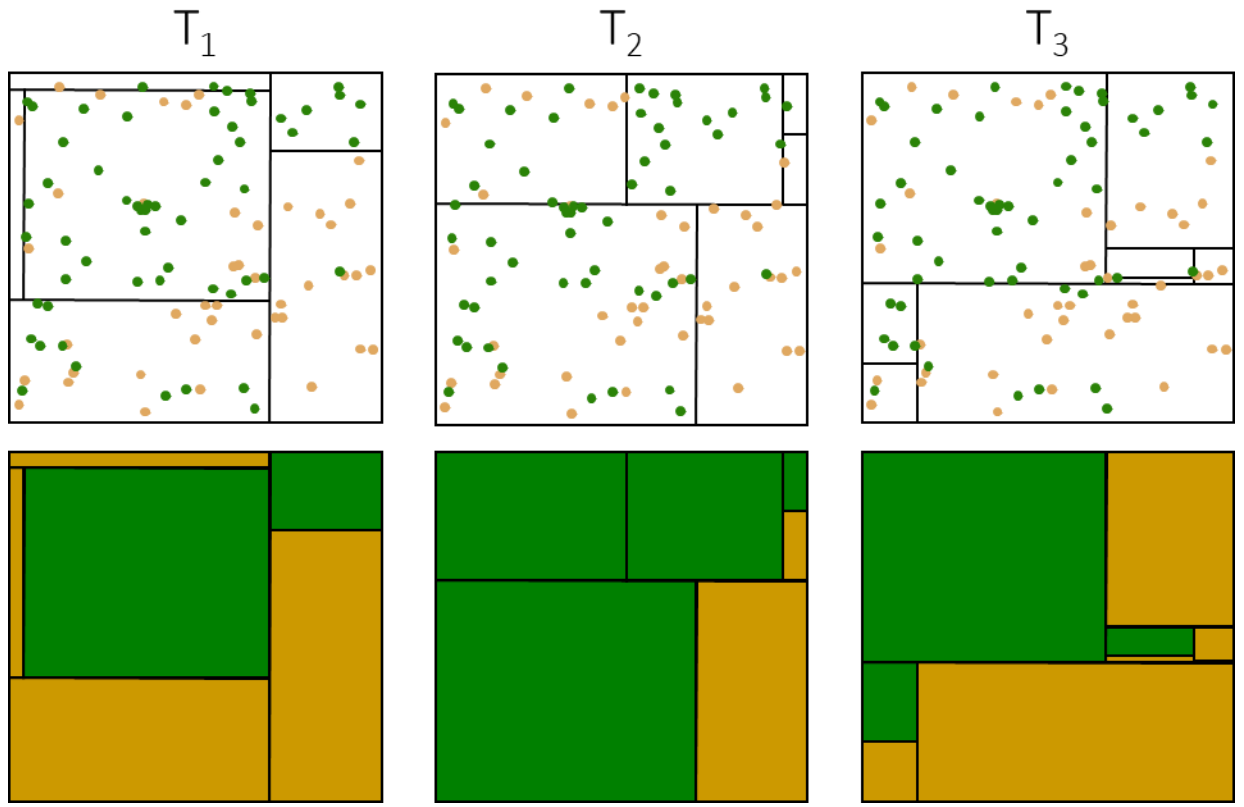
Perturb And Combine Methods

- Perturb and combine methods generate multiple models by manipulating the distribution of the data or altering the construction method.
- Any unstable modelling method can use perturb and combine methods.



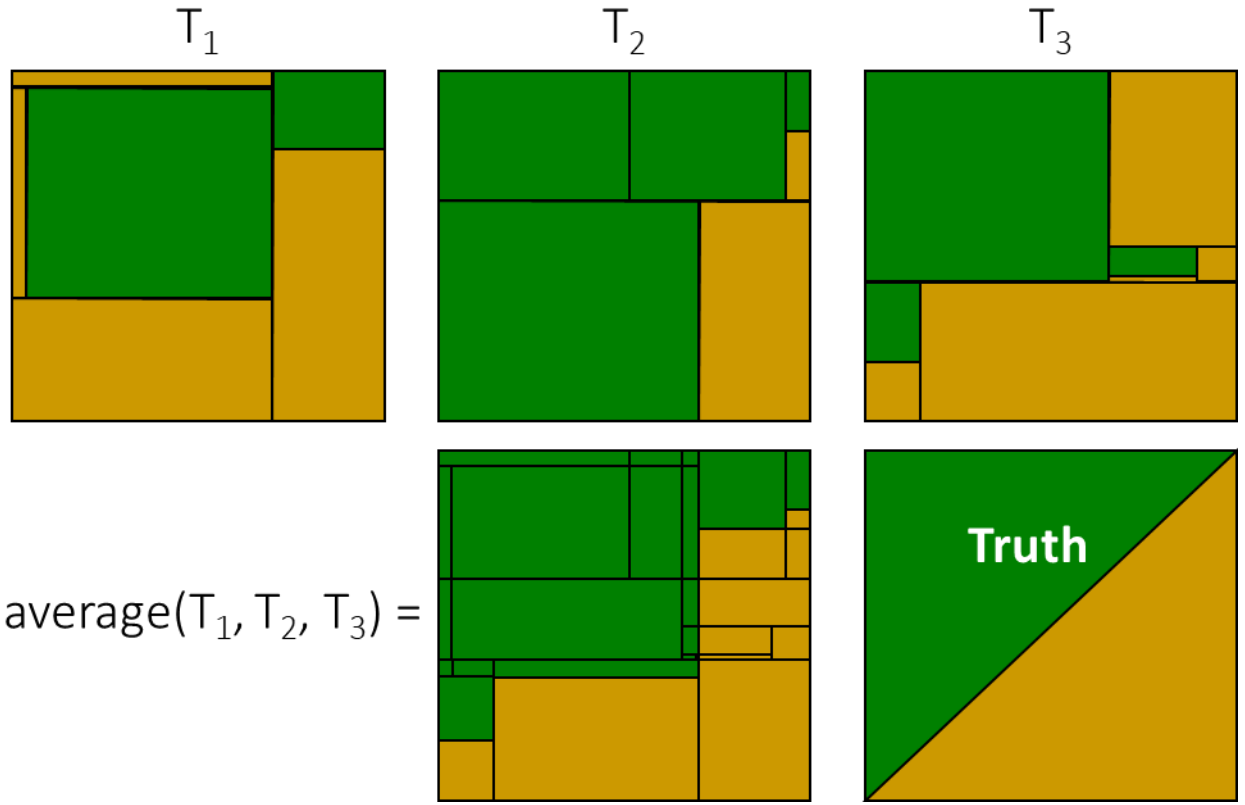
Perturb And Combine Methods

- Perturb and combine methods take advantage of the instability of decision trees to create models that are more powerful.
- Some perturbation methods:
  - resample
  - subsample
  - add noise
  - adaptively reweight
  - randomly choose from the competitor splits.



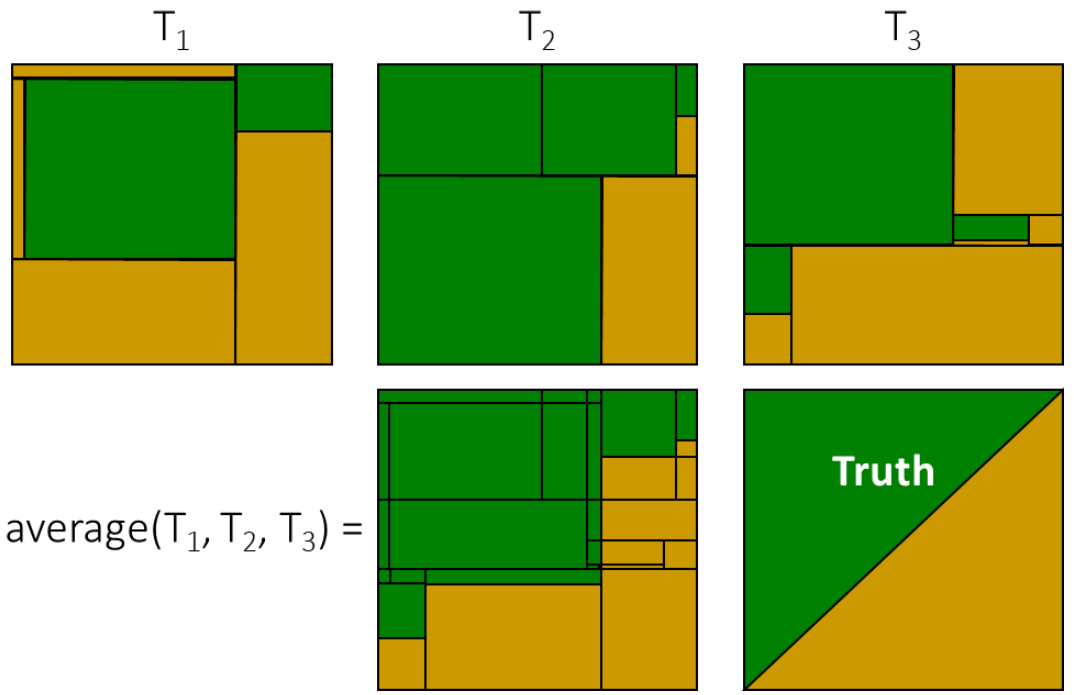
Perturb And Combine Methods

- An ensemble model is the combination of multiple models.
- The combinations methods:
  - voting on the classifications
  - weighted voting where some models have more weight
  - averaging (weighted or unweighted) the predicted values.



Perturb And Combine Methods

- Perturb and combine methods can improve models' performance by variance reduction.
- But combing stable models can negatively affect performance.
- A new case is scored by running it down the multiple trees and averaging the results.
- So multiple models need to be stored and processed, and the interpretation of single tree is lost.

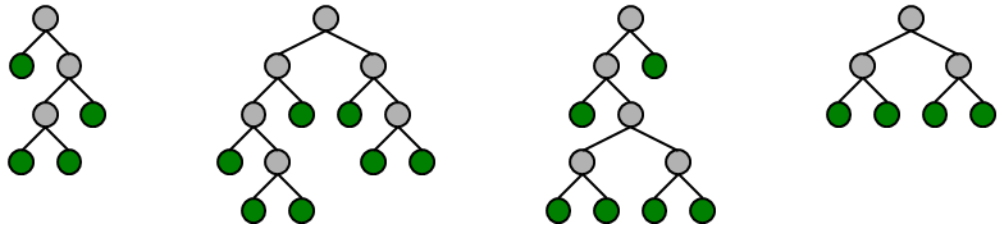


Bagging, Boosting and Gradient Boosting

Bagging (Bootstrap Aggregation)

1. Draw *K* bootstrap samples.
   A random sample of size *n* is drawn with replacement from the empirical distribution of a sample of size *n*.
2. Build a tree on each bootstrap sample.
   Large decision trees with low bias and high variance are ideal.

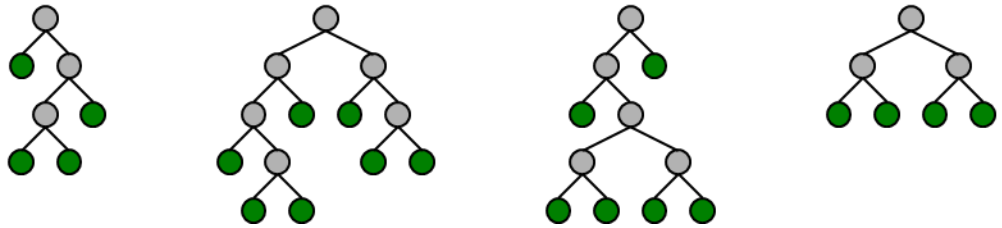
	k=1	k=2	k=3	k=4 …
case	freq	freq	freq	freq
1	1	0	3	1
2	0	1	1	1
3	2	0	0	2
4	0	2	2	0
5	2	2	0	1
6	1	1	0	1



Bagging (Bootstrap Aggregation)

3. Vote or average.
   The mean of the posterior probabilities or the plurality vote of the predicted class is taken for classification problems.
   For the regression problem, the mean of the predicted values is taken.

	k=1	k=2	k=3	k=4 …
case	freq	freq	freq	freq
1	1	0	3	1
2	0	1	1	1
3	2	0	0	2
4	0	2	2	0
5	2	2	0	1
6	1	1	0	1



Boosting

- Boosting is machine learning ensemble algorithm for mainly reducing bias and variance.
- Arcing (Adaptive resampling and combining) methods are examples of boosting

	k=1		k=2		k=3		k=4 …
case	freq	m	freq	m	freq	m	freq
1	1	1	1.5	1	.5	2	.97
2	1	0	.75	0	.25	0	.06
3	1	1	1.5	2	4.25	3	4.69
4	1	0	.75	1	.5	1	.11
5	1	0	.75	0	.25	0	.06
6	1	0	.75	0	.25	1	.11



*Shown is Arc-x4, one method of boosting

Boosting

- For Arc-x4, at the kth step, a decision tree model is fit using weights for each case.
- For the ith case, the Arc-x4 weights are as follows:

$$p(i) = \frac{1 + m(i)^4}{\sum(1 + m(i)^4)}$$

where $0 \leq m(i) \leq k$ is the number of times that the ith case is misclassified in the preceding steps

	k=1		k=2		k=3		k=4 …
case	freq	m	freq	m	freq	m	freq
1	1	1	1.5	1	.5	2	.97
2	1	0	.75	0	.25	0	.06
3	1	1	1.5	2	4.25	3	4.69
4	1	0	.75	1	.5	1	.11
5	1	0	.75	0	.25	0	.06
6	1	0	.75	0	.25	1	.11



*Shown is Arc-x4, one method of boosting

Boosting

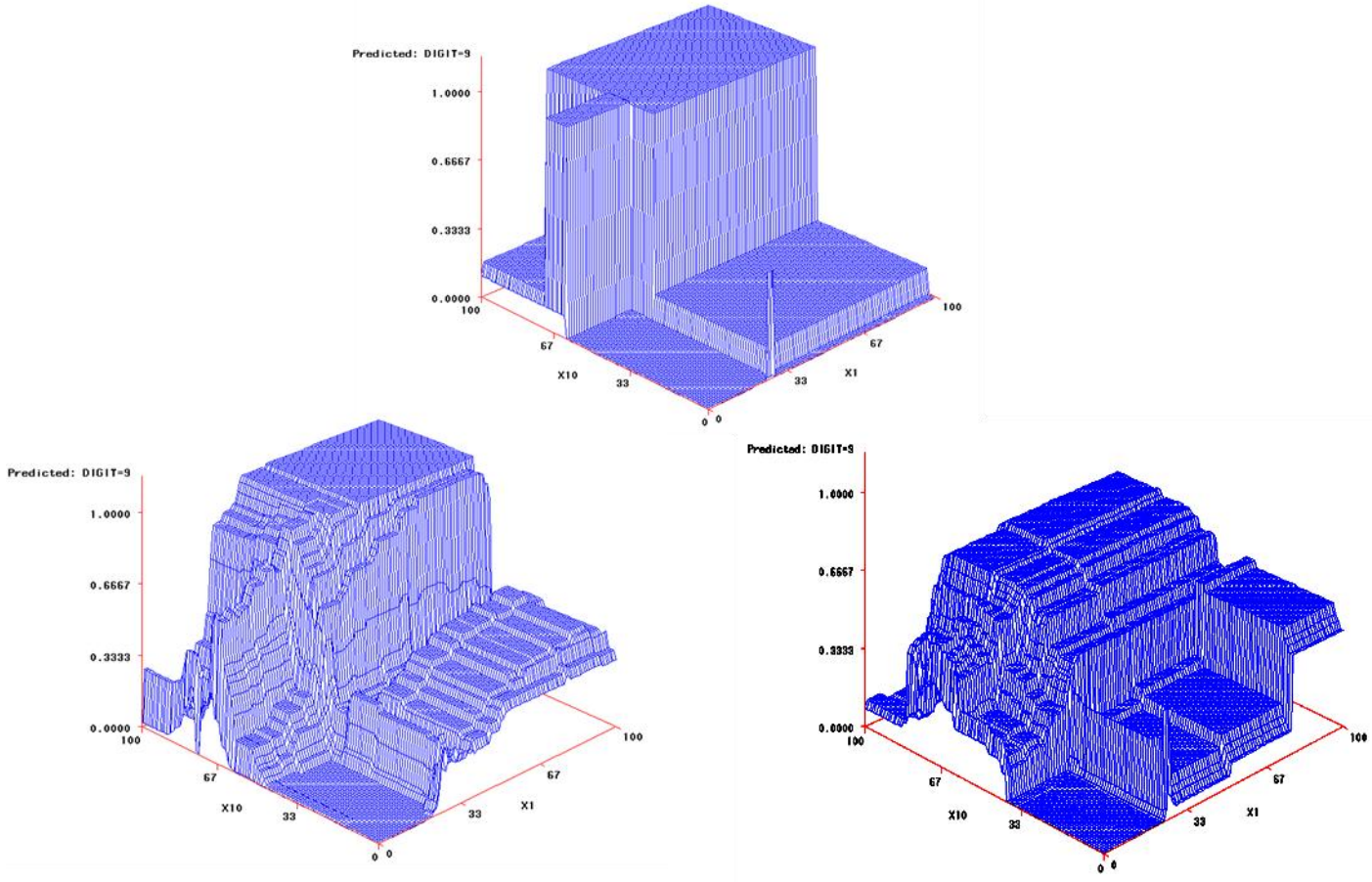
- Pruning the individual tree improves boosting performance.
- The process is repeated *K* times, and the *K* models are combined by voting or averaging the posterior probabilities.

	k=1		k=2		k=3		k=4 …
case	freq	m	freq	m	freq	m	freq
1	1	1	1.5	1	.5	2	.97
2	1	0	.75	0	.25	0	.06
3	1	1	1.5	2	4.25	3	4.69
4	1	0	.75	1	.5	1	.11
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*Shown is Arc-x4, one method of boosting

Single, Bagged and Boosted Decision Tree



Gradient Boosting with Decision Trees

- The gradient boosting algorithm is similar to standard boosting, except at each iteration, the target is the residual from the previous decision tree model.
- At each step, the accuracy of the tree is computed.
- Successive samples are adjusted to accommodate previous inaccuracies.
- The model is a weighted ($\beta_1 \ldots \beta_M$) linear combination of (usually) simple models.

Forest Models

Forest Models

- A forest model is an ensemble of classification or regression trees.
- Trees in the forest differ from each other in two ways.
  - Training data for a tree is a sample without replacement from all observations.
  - Input variables considered for splitting a node are randomly selected from available inputs. Only the variable most associated with the target is split for that node.

Forest Models

- The forest algorithm samples the rows and the columns at each step.
- The forest algorithm perturbs the training data more than the bagging algorithm.
- This increased variation among the trees in the ensemble often leads to improved predictive accuracy.

Out-of-Bag Sample

- The out-of-bag sample refers to the training data that is excluded during the construction of an individual tree.
- Observations in the training data that are used to construct an individual tree are the bagged sample.
- Some model assessments such as the iteration plots are computed using the out-of-bag sample as well as all the training data.

Forest Models

- Trees automatically handle missing values and variable reduction. Therefore, the input data requires less preparation.
- Forests tend to give better prediction than any specific tree, and often outperform other classes of models.
- Forests are challenging to interpret, but they can be considered an “ideal” model for other models to be compared against.

Lesson Summary

- Ensemble models take advantage of the inherent instability of decision trees.
- Bagging resamples training data with replacement and then fits models to each sample.
- Boosting adaptively reweights subsequent samples based on mistakes that are made in target classification from previous samples, and it can be applied only to categorical targets
- The Gradient boosting algorithm is a weighted linear combination of simple models. Subsequent models are fit to residuals from previous combinations of models.
- Forest models sample both rows and columns of the data for greater diversity in the trees to improve predictive accuracy.

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Can we use SAS Viya to implement Gradient Boosting and Forest algorithm?



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